Baryon Asymmetry With The CP Violation

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Abstract

This thesis report presents a theoretical study of baryon number violation in the Standard Model, focusing on its quantum and topological origins. Baryon number is conserved at the classical level, but it is violated at the quantum level due to chiral anomalies linked to electroweak gauge fields. The report discusses the role of the Higgs field topology, the Chern-Simons number, and non-perturbative vacuum transitions in detail. It also looks at CP violation from the CKM matrix, which is the only source of CP violation in the Standard Model. These aspects are examined in the context of early-universe cosmology and the possible creation of baryon asymmetry.

Keywords: Electroweak Theory, Baryon Number Violation, Anomalies, Topology, Effective Potential, CP Violation, CKM Matrix

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Chapter 1

Introduction

One of the deepest puzzles in modern cosmology is why the observable universe appears to be made almost entirely of matter, with no significant reservoir of primordial antimatter detected at any astrophysical scale. If the laws of physics treated matter and antimatter symmetrically, one would not expect this situation to arise naturally. Yet it evidently did, at some epoch in the early universe.

At early times, the universe was a hot, dense plasma in which creation and annihilation processes such as

$$e^- + e^+ \longleftrightarrow \gamma + \gamma \tag{1.1}$$

proceeded rapidly in both directions. Thermal equilibrium ensured that particles and antiparticles were produced at comparable rates, so without some explicit symmetry-breaking mechanism one would expect

$$n_B = n_{\bar{B}}, \tag{1.2}$$

where n_B and $n_{\bar{B}}$ are the baryon and antibaryon number densities respectively.

The departure from this symmetric picture is quantified by the baryon-to-photon ratio

$$\eta_B = \frac{n_B - n_{\bar{B}}}{n_\gamma}, \tag{1.3}$$

which measurements pin at

$$\eta_B \sim 10^{-10}. \tag{1.4}$$

Despite being a tiny number, this asymmetry is large enough to account for all the baryonic matter observed today. Explaining its origin from first principles is the central task of baryogenesis [39].

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1.1 Sakharov Conditions

The conditions under which a baryon asymmetry can be dynamically generated were first laid out by Sakharov in 1967. Three requirements must be simultaneously satisfied; they are now universally referred to as the Sakharov conditions.

1.1.1 Violation of Baryon Number

Baryon number violation is a prerequisite: a universe that starts with $B = 0$ and respects an exact baryon-number symmetry at every subsequent moment cannot develop any net baryon excess. Processes that change B must therefore exist. The net baryon number is defined as

$$\Delta B = N_B - N_{\bar{B}}, \tag{1.5}$$

where N_B and $N_{\bar{B}}$ are the total numbers of baryons and antibaryons. For baryogenesis to occur, one must have

$$\Delta B \neq 0. \tag{1.6}$$

At the quark level, each quark carries baryon number

$$B = \frac{1}{3}, \tag{1.7}$$

while each antiquark carries

$$B = -\frac{1}{3}. \tag{1.8}$$

Hence, for any state,

$$B = \frac{1}{3} \sum_q \left(n_q - n_{\bar{q}} \right). \tag{1.9}$$

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1.1.2 Violation of C and CP Symmetry

Charge conjugation symmetry C transforms particles into antiparticles. If C symmetry were exact, matter and antimatter would be produced at identical rates. Similarly, charge-parity symmetry CP combines charge conjugation and spatial inversion. If CP were exact, baryon-number-violating interactions would still generate equal amounts of matter and antimatter.

Within the Standard Model, weak interactions violate C symmetry. Moreover, CP violation arises from the complex phase of the Cabibbo-Kobayashi-Maskawa (CKM) quark mixing matrix [8, 40]. A basis-independent quantitative measure of this violation is given by the Jarlskog invariant [23, 28]. However, the amount of Standard Model CP violation is generally considered too small to explain the observed baryon asymmetry.

1.1.3 Departure from Thermal Equilibrium

In thermal equilibrium, any generated asymmetry tends to be erased by inverse reactions. Therefore, baryogenesis requires nonequilibrium conditions. Such conditions may arise during the expansion of the early universe or during cosmological phase transitions [39].

1.2 Baryon Number Violation in the Standard Model

At the perturbative level, the Standard Model conserves baryon number. However, non-perturbative electroweak effects violate $B + L$ while conserving $B - L$. These transitions are associated with topologically distinct vacuum sectors of the electroweak gauge theory.

At sufficiently high temperature, sphaleron transitions become unsuppressed and can efficiently change baryon number. Consequently, they play a central role in electroweak baryogenesis and in leptogenesis, where a primordial lepton asymmetry is partially converted into baryon asymmetry [39].

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1.3 Models of Baryogenesis

Several theoretical mechanisms have been proposed to explain the origin of the observed baryon asymmetry.

1.3.1 GUT Baryogenesis

In grand unified theories, baryon number violation may occur through the decay of superheavy gauge or Higgs bosons. If these decays violate CP and occur out of equilibrium, they can generate a net baryon asymmetry.

1.3.2 Electroweak Baryogenesis

Electroweak baryogenesis operates near the electroweak scale. It relies on sphaleron-induced baryon number violation, sufficient CP violation, and a strongly first-order electroweak phase transition. Under these conditions, the asymmetry may be generated during the electroweak epoch [39].

1.3.3 Leptogenesis

In leptogenesis, a lepton asymmetry is first generated, commonly through the decay of heavy Majorana neutrinos. Electroweak sphalerons subsequently convert part of this

lepton asymmetry into baryon asymmetry.

1.4 Motivation and Scope of the Present Work

The question of why our universe has more matter than empty radiation is one of the biggest puzzles in modern cosmology. At first, the Standard Model of particle physics seems well-suited to address this issue. It naturally includes all three of Sakharov's key conditions for baryogenesis. Baryon number is violated through high-temperature sphaleron transitions. The complex phase in the CKM matrix breaks C and CP symmetries. The expanding early universe created a break from thermal equilibrium during the electroweak epoch. Given that these mechanisms are fundamental to the theory, we must ask whether the Standard Model can explain the observed matter-antimatter asymmetry.

However, detailed studies reveal serious theoretical challenges within this framework. First, the amount of CP violation from the quark sector is limited by the small quark masses and mixing angles. This leads to an asymmetry that falls many orders of magnitude short of what observations require. Second, with the Higgs boson mass measured around 125 GeV, we find that the electroweak phase transition is a smooth thermal crossover rather than a strong first-order process. Therefore, even if a sufficient asymmetry did occur, it would quickly be wiped out by unimpeded sphaleron interactions.

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These significant numerical issues motivate this work. This report aims to carefully and quantitatively reevaluate baryogenesis solely within the Standard Model [39, 24]. By calculating the theoretical baryon-to-entropy ratio and evaluating the limits of CP violation, this work seeks to clearly demonstrate the extent of the Standard Model's shortcomings. This will emphasize the critical need for new physics beyond it.

Chapter 2

Baryon Asymmetry In Standard Model

Electroweak baryogenesis (EWBG) is a theoretical framework in which all three Sakharov conditions are satisfied within or near the electroweak scale through a combination of anomalous baryon number violating processes and electroweak phase-transition dynamics.
In this scenario, the observed matter–antimatter asymmetry of the universe is generated during the electroweak epoch, when the temperature of the universe was of order 100 GeV.

In this chapter, we will calculate the baryon to entropy densities ratio resulting from the CP transformation of the effective action at the electroweak scale.

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2.1 Baryon Number Violation In Electroweak Theory

The Standard Model of unified electromagnetic and weak nuclear interactions is based on the gauge group $SU(2)_L \times U(1)_Y$. This theory is described by the Lagrangian density[39]

$$\mathcal{L} = (D_\mu \phi)^\dagger (D^\mu \phi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} W^a_{\mu\nu} W^{a\,\mu\nu} + V(\phi) + \mathcal{L}_f. \tag{2.1}$$

Here, the field strength tensors are defined as

$$F_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu, \tag{2.2}$$

$$W^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, \epsilon^{abc} A^b_\mu A^c_\nu \tag{2.3}$$

where B_μ is the hypercharge gauge field and A^a_μ are the weak isospin gauge fields.

The covariant derivative acting on the Higgs field is given by

$$D_\mu = \partial_\mu - \frac{i}{2} g\, \boldsymbol{\tau} \cdot \boldsymbol{A}_\mu - \frac{i}{2} g'\, B_\mu \tag{2.4}$$

where $\boldsymbol{\tau}$ are the Pauli matrices.

The Higgs potential takes the form[39]

$$V(\phi) = \frac{\lambda}{4}\left(\phi^{\dagger}\phi - v^2\right)^2. \tag{2.5}$$

In the above expressions, g is the $SU(2)$ coupling constant, g' is the $U(1)$ coupling constant, λ is the Higgs self-coupling, and

$$v = 246 \text{ GeV}$$

is the vacuum expectation value (VEV) of the Higgs field.

To deal with the baryogenesis,it is convenient to rewrite equation (2.1) in such a way that $\mathcal{L}_f$ and U(1) hypercharge gauge field are ignored and we use the matrix notations;

$$\mathcal{L} = -\frac{1}{2}\,\mathrm{Tr}\big(W_{\mu\nu}W^{\mu\nu}\big) - \frac{1}{2}\,\mathrm{Tr}(D^{\mu}\Phi)^{\dagger}\,D_{\mu}\Phi - \frac{\lambda}{4}\Big[\,\mathrm{Tr}\Big(\Phi^{\dagger}\Phi\Big) - v^2\Big]^2.$$

and now

$$D_{\mu} = \partial_{\mu} - \frac{i}{2}g\,\boldsymbol{\tau}\cdot\boldsymbol{A}_{\mu}$$

We define the matrix-valued field as

$$W_{\mu\nu} = \frac{1}{2}\tau^a W^a_{\mu\nu},$$

Using the Pauli matrix identity

$$\mathrm{Tr}\Big(\tau^a\tau^b\Big) = 2\,\delta^{ab},$$

one finds

$$\mathrm{Tr}\big(W_{\mu\nu}W^{\mu\nu}\big) = \frac{1}{4}\,\mathrm{Tr}\Big(\tau^a\tau^b\Big)\,W^a_{\mu\nu}W^{b\,\mu\nu} = \frac{1}{2}\,W^a_{\mu\nu}W^{a\,\mu\nu}.$$

so that

$$-\frac{1}{4}\,W^a_{\mu\nu}W^{a\,\mu\nu} = -\frac{1}{2}\,\mathrm{Tr}\big(W_{\mu\nu}W^{\mu\nu}\big)\,.$$

In this representation, the usual Higgs doublet[39] $\phi = (\phi_1, \phi_2)$ is expressed in terms of a 2×2 matrix field Φ as :

$$\Phi(x,t) = \begin{pmatrix} \phi_2^* & \phi_1 \\ -\phi_1^* & \phi_2 \end{pmatrix}.$$

its Hermitian conjugate is given by

$$\Phi^{\dagger} = \begin{pmatrix} \phi_2 & -\phi_1 \\ \phi_1^* & \phi_2^* \end{pmatrix}.$$

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Now ;

$$\Phi^\dagger\Phi = \begin{pmatrix} \phi_2 & -\phi_1 \\ \phi_1^* & \phi_2^* \end{pmatrix} \begin{pmatrix} \phi_2^* & \phi_1 \\ -\phi_1^* & \phi_2 \end{pmatrix}.$$

the product simplifies to

$$\Phi^\dagger\Phi = (\phi_1^*\phi_1 + \phi_2^*\phi_2) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Taking the trace, we obtain

$$\mathrm{Tr}\left(\Phi^\dagger\Phi\right) = 2\left(\phi_1^*\phi_1 + \phi_2^*\phi_2\right).$$

defining the real scalar field σ by

$$\sigma^2 \equiv 2\left(\phi_1^*\phi_1 + \phi_2^*\phi_2\right) = \mathrm{Tr}\left(\Phi^\dagger\Phi\right).$$

Since $\Phi^\dagger\Phi$ is proportional to the identity matrix, the matrix field Φ can be factorized into a radial part and a unitary part as[39]

$$\Phi = \frac{\sigma}{\sqrt{2}}\,\mathrm{U},$$

where $U \in \mathrm{SU}(2)$ is a unitary matrix with $\det U = 1$, defined uniquely at each spacetime point where $\sigma \neq 0$.

and U is an $SU(2)$-valued field that is uniquely defined at every spacetime point where σ does not vanish.

Φ is a matrix valued field whose norm is fixed by σ but its orientation is unconstrained.

Therefore we can write simply: [39]

$Higgsfield = magnitude \times orientation$

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We impose the conditions that at all times t ;

$$\lim_{|\mathbf{x}|\to\infty} \sigma(\mathbf{x}, t) = v,$$

and

$$\lim_{|\mathbf{x}|\to\infty} U(\mathbf{x}, t) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

In the $A_0 = 0$ gauge, a vacuum configuration is given by

$$\Phi = \frac{v}{\sqrt{2}}\, U, \qquad \textit{and}, A_j = \frac{1}{ig}\, \partial_j U\, U^\dagger.$$

So we can write $U(\mathbf{x}, t)$ can be associated with an integer-valued winding number,

$$N_H(t) \equiv w[U] = \frac{1}{24\pi^2} \int d^3x\, \epsilon^{ijk}\, \mathrm{Tr}\Big(U^\dagger \partial_i U\, U^\dagger \partial_j U\, U^\dagger \partial_k U \Big)\,,$$

known as the *Higgs winding number*.

If the Higgs field $\Phi(\mathbf{x}, t)$ evolves continuously in time, the quantity $N_H(t)$ can change only at times when σ vanishes at some point in space. At such times, N_H is not defined; at all other times, it takes integer values.

It is important to note that the Higgs winding number of a vacuum configuration is equal to its *Chern–Simons number* [39]

$$N_{CS}(t) \equiv \frac{g^2}{32\pi^2} \int d^3x\, \epsilon^{ijk}\, \mathrm{Tr}\Big(A_i \partial_j A_k + \frac{2}{3} ig\, A_i A_j A_k \Big)\,.$$

For a general non-vacuum configuration, however, the Chern–Simons number is not integer-valued.

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At the classical level, the number of each fermionic species is conserved independently in all physical processes. This conservation follows from the existence of a global $U(1)$ symmetry associated with each fermion species. In the Standard Model, there are 12 fermionic species, and consequently twelve separate classically conserved global currents.[39]

In particular, the baryon number current can be written in a vector-like form as

$$j_B^\mu = \frac{1}{2}\, \bar{Q} \gamma^\mu Q,$$

where Q denotes the quark fields, with sum over color and flavor indices.

However, quantum effects significantly modify this classical picture. In a gauge theory, any axial current of a gauge-coupled Dirac fermion,

$$\bar{\psi} \gamma^\mu \gamma^5 \psi,$$

is anomalous at the quantum level. This fact becomes crucial for baryon number conservation because electroweak fermions couple chirally to the gauge fields.

To make this explicit, the baryon current may be decomposed into left-handed and right-handed components as[24]

$$j_B^\mu = \frac{1}{4}\left[\bar{Q}\gamma^\mu(1-\gamma^5)Q + \bar{Q}\gamma^\mu(1+\gamma^5)Q\right].$$

When computing the divergence of this current, only the axial part contributes to the anomalous violation of baryon number, while the purely vector part remains conserved.

This quantum effect can be understood by non-vanishing divergences of the baryon and lepton number currents. As a result, one finds[24]

$$\partial_\mu j_B^\mu = \partial_\mu j_L^\mu = n_f\left(\frac{g^2}{32\pi^2}W^a_{\mu\nu}\tilde{W}^{a\,\mu\nu} - \frac{g'^2}{32\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}\right),$$

where n_f denotes the number of fermion families.The dual of the $SU(2)$ field strength tensor is defined as

$$\tilde{W}^{\mu\nu} = \frac{1}{2}\,\epsilon^{\mu\nu\alpha\beta}W_{\alpha\beta},$$

and analogously the dual of the $U(1)$ field strength tensor is

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\,\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}.$$

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The importance of this anomaly is closely tied to the non-trivial vacuum structure of the electroweak theory. The above expression for the divergence may be rewritten as a total derivative,

$$\partial_\mu j_B^\mu = \partial_\mu j_L^\mu = n_f\left(\frac{g^2}{32\pi^2}\,\partial_\mu K^\mu - \frac{g'^2}{32\pi^2}\,\partial_\mu k^\mu\right),$$

where the gauge non-invariant currents are defined by[24]

$$K^\mu = \epsilon^{\mu\nu\alpha\beta}\left(W^a_\nu\partial_\alpha W^a_\beta - \frac{1}{3}g\,\epsilon^{abc}W^a_\nu W^b_\alpha W^c_\beta\right),$$

$$k^\mu = \epsilon^{\mu\nu\alpha\beta}F_{\nu\alpha}B_\beta.$$

Relation Between Baryon Number Change and the Chern–Simons Number

The divergence of the baryon current is given by[24]

$$\partial_\mu j_B^\mu = n_f\left(\frac{g^2}{32\pi^2}W^a_{\mu\nu}\tilde{W}^{a\mu\nu} - \frac{g'^2}{32\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}\right) \tag{2.6}$$

For vacuum-to-vacuum transitions the $U(1)$ contribution does not lead to a change in baryon number, since the spacetime integral of $F_{\mu\nu}\tilde{F}^{\mu\nu}$ vanishes. Therefore we retain only the $SU(2)$ contribution:

$$\partial_\mu j_B^\mu = n_f \frac{g^2}{32\pi^2} W^a_{\mu\nu} \tilde{W}^{a\mu\nu}. \tag{2.7}$$

The total baryon number(by the Noether theorem the conserved baryonic charge) is defined as :[24]

$$B(t) = \int d^3x \, j_B^0(\mathbf{x}, t). \tag{2.8}$$

The change in baryon number between an initial time $t = 0$ and a final time $t = t_f$ is

$$\Delta B = B(t_f) - B(0) = \int d^3x \left[j_B^0(x, t_f) - j_B^0(x, 0) \right].$$

Using

$$j_B^0(x, t_f) - j_B^0(x, 0) = \int_0^{t_f} dt \, \partial_0 j_B^0(x, t),$$

we obtain

$$\Delta B = \int_0^{t_f} dt \int d^3x \, \partial_0 j_B^0.$$

From the divergence of the current

$$\partial_\mu j_B^\mu = \partial_0 j_B^0 + \partial_i j_B^i,$$

we write

$$\partial_0 j_B^0 = \partial_\mu j_B^\mu - \partial_i j_B^i.$$

Substituting gives

$$\Delta B = \int_0^{t_f} dt \int d^3x \left(\partial_\mu j_B^\mu - \partial_i j_B^i \right).$$

The spatial divergence term becomes a surface term,

$$\int d^3x \, \partial_i j_B^i = \int_{\partial V} dS_i \, j_B^i,$$

which vanishes for vacuum configurations where the current vanishes at spatial infinity. Therefore[24]

$$\Delta B = \int_0^{t_f} dt \int d^3x \, \partial_\mu j_B^\mu.$$

Substituting the anomaly equation gives

$$\Delta B = n_f \frac{g^2}{32\pi^2} \int_0^{t_f} dt \int d^3x \, W^a_{\mu\nu} \tilde{W}^{a\mu\nu}. \tag{2.9}$$

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For non-Abelian gauge fields the following identity holds:

$$W^a_{\mu\nu}\tilde{W}^{a\mu\nu} = \partial_\mu K^\mu, \tag{2.10}$$

where K^μ is the Chern–Simons current. Using this relation we obtain

$$\Delta B = n_f \frac{g^2}{32\pi^2} \int d^4x\, \partial_\mu K^\mu. \tag{2.11}$$

Applying Gauss's theorem converts the spacetime integral of the divergence into boundary terms:

$$\int d^4x\, \partial_\mu K^\mu = \int d^3x\, K^0(t_f) - \int d^3x\, K^0(0). \tag{2.12}$$

The Chern–Simons number is defined as[24]

$$N_{CS}(t) = \frac{g^2}{32\pi^2} \int d^3x\, K^0(\mathbf{x}, t). \tag{2.13}$$

Substituting this definition into the previous expression yields

$$\Delta B = n_f \left[N_{CS}(t_f) - N_{CS}(0) \right]. \tag{2.14}$$

Thus, the change in baryon number is directly related to the change in the Chern–Simons number:[39]

$$\boxed{\Delta B = n_f\, \Delta N_{CS}} \tag{2.15}$$

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This result shows that baryon number violation in the electroweak theory is associated with changes in the topological structure of the $SU(2)$ gauge fields. Transitions between vacua with different Chern–Simons numbers can occur through non-perturbative processes such as instantons and sphalerons.[24]

2.2 Classical Baryon Number Conservation

At the classical level, baryon number conservation arises from a global $U(1)$ symmetry of the fermionic Lagrangian. The baryon number current for quarks is given by

$$j^\mu_B = \frac{1}{2}\, \bar{Q}\gamma^\mu Q, \tag{2.16}$$

which satisfies

$$\partial_\mu j^\mu_B = 0 \tag{2.17}$$

where Q represents quarks, and the sum is taken over colour and flavor indices. When classical equations of motion are used. This is a standard Vector Current. In classical

physics, $\partial_\mu j^\mu = 0$, meaning the total Baryon number is conserved. We do not see protons decaying into other random particles or popping out of nowhere.

2.3 Expectation Value of the Divergence of the Baryon Current

In order to study baryon number violation in the quantum theory, it is convenient to evaluate the expectation value of the divergence of the baryon current using the path integral formalism [31]. In quantum field theory the expectation value of any operator $\mathcal{O}$ is given by the functional integral over all field configurations weighted by the exponential of the action [34].[39]

For any operator $\mathcal{O}$, this can be written as

$$\langle \mathcal{O} \rangle = \int \mathcal{D}\Phi \, \mathcal{O}[\Phi] \, \exp(iS[\Phi]) \,, \tag{2.18}$$

where $S = \int d^4x \, \mathcal{L}$ is the action of field Φ representing all dynamical fields.[39]

Applying this definition to the divergence of the baryon current $\partial_\mu j_B^\mu$, the quantum expectation value becomes [20]

$$\langle \partial_\mu j_B^\mu \rangle = \int dA_\mu^a \, d\bar{\Psi} \, d\Psi \, dh \, \partial_\mu j_B^\mu \exp\left[i \int d^4x \, \mathcal{L} \right] . \tag{2.19}$$

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Here Ψ and $\bar{\Psi}$ denote all fermionic fields in the theory, A_μ^a represents the gauge boson fields, and h denotes the scalar (Higgs) fields. [39]

Using the definition of the baryon current (from eqn 2.16)

$$j_B^\mu = \frac{1}{2} \bar{\Psi} \gamma^\mu \Psi, \tag{2.20}$$

its divergence can be written as

$$\partial_\mu j_B^\mu = \frac{1}{2} \left[(\partial_\mu \bar{\Psi}) \gamma^\mu \Psi + \bar{\Psi} \gamma^\mu (\partial_\mu \Psi) \right] . \tag{2.21}$$

Taking the divergence of j_B^μ gives

$$\partial_\mu j_B^\mu = \frac{1}{2} \left[(\partial_\mu \bar{\Psi}) \gamma^\mu \Psi + \bar{\Psi} \gamma^\mu (\partial_\mu \Psi) \right] .$$

Consider the Dirac action for fermions

$$S = \int d^4x \, \bar{\Psi} (i\gamma^\mu \partial_\mu - m) \Psi .$$

Varying the action with respect to $\bar{\Psi}$ gives

$$\delta S = \int d^4x\, \delta\bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi.$$

Therefore

$$\frac{\delta S}{\delta\bar{\Psi}} = (i\gamma^\mu\partial_\mu - m)\Psi.$$

Next vary the action with respect to Ψ

$$\delta S = \int d^4x\, \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\delta\Psi.$$

Integrating the kinetic term by parts,

$$\delta S = -\int d^4x\, (\partial_\mu\bar{\Psi})i\gamma^\mu\delta\Psi - \int d^4x\, m\bar{\Psi}\delta\Psi.$$

Thus

$$\frac{\delta S}{\delta\Psi} = -i\partial_\mu\bar{\Psi}\gamma^\mu - m\bar{\Psi}.$$

Using these relations we compute

$$i\bar{\Psi}\frac{\delta S}{\delta\bar{\Psi}} = i\bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi = -\bar{\Psi}\gamma^\mu\partial_\mu\Psi - im\bar{\Psi}\Psi$$

and

$$i\frac{\delta S}{\delta\Psi}\Psi = i(-i\partial_\mu\bar{\Psi}\gamma^\mu - m\bar{\Psi})\Psi = (\partial_\mu\bar{\Psi})\gamma^\mu\Psi - im\bar{\Psi}\Psi.$$

Adding the two expressions gives

$$i\bar{\Psi}\frac{\delta S}{\delta\bar{\Psi}} + i\frac{\delta S}{\delta\Psi}\Psi = (\partial_\mu\bar{\Psi})\gamma^\mu\Psi - \bar{\Psi}\gamma^\mu\partial_\mu\Psi - 2im\bar{\Psi}\Psi.$$

Since the baryon current corresponds to a vector symmetry $\Psi \to e^{i\alpha}\Psi$, the mass term is invariant and does not contribute to the divergence. Hence the divergence of the current can be written as[39]

$$\partial_\mu j_B^\mu = \frac{1}{2}\left[i\bar{\Psi}\frac{\delta S}{\delta\bar{\Psi}} + i\frac{\delta S}{\delta\Psi}\Psi\right]$$

So now ,we can write eqn (2.19) as :

$$\langle\partial_\mu j_B^\mu\rangle = \frac{1}{2}\int dA_\mu^a\, d\bar{\Psi}\, d\Psi\, dh\left[i\bar{\Psi}\frac{\delta S}{\delta\bar{\Psi}} + i\frac{\delta S}{\delta\Psi}\Psi\right]\exp\left[i\int d^4x\,\mathcal{L}\right]$$

Using the Schwinger–Dyson identity for Green's functions derived in Peskin and Schroeder (Eq. 9.88), one has for a generic field ϕ

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$$\left\langle \frac{\delta S}{\delta \phi(x)} \phi(x_1) \cdots \phi(x_n) \right\rangle = \sum_{i=1}^{n} \langle \phi(x_1) \cdots (i\delta(x - x_i)) \cdots \phi(x_n) \rangle .$$

Applying the Schwinger–Dyson relation with $n = 1$ and $\phi(x_1) = \Psi(x)$ gives

$$\left\langle \frac{\delta S}{\delta \Psi(x)} \Psi(x) \right\rangle = i\delta(0).$$

Multiplying both sides by i gives

$$\left\langle i\frac{\delta S}{\delta \Psi(x)} \Psi(x) \right\rangle = -\delta(0).$$

Next we apply the same identity to $\bar{\Psi}$

$$\left\langle \bar{\Psi}(x) \frac{\delta S}{\delta \bar{\Psi}(x)} \right\rangle = i\delta(0).$$

Multiplying again by i yields

$$\left\langle i\bar{\Psi}(x) \frac{\delta S}{\delta \bar{\Psi}(x)} \right\rangle = -\delta(0).$$

Substituting these results into the path integral expression gives

$$\begin{aligned} \langle \partial_\mu j_B^\mu \rangle &= \frac{1}{2} \int dA_\mu^a \, d\bar{\Psi} \, d\Psi \, dh \left[\delta A_\mu^a - \delta A(0) \right] \exp\left[i \int d^4x \, \mathcal{L} \right] \\ &= \delta A(0) \int dA_\mu^a \, d\bar{\Psi} \, d\Psi \, dh \exp\left[i \int d^4x \, \mathcal{L} \right] . \end{aligned} \tag{2.22}$$

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Finally we sum over all fermion species in the theory. In the Standard Model the total number of fermionic degrees of freedom contributing to the baryon current is 48, which gives[39]

$$\langle \partial_\mu j_B^\mu \rangle = 24 \, \delta(0) \int dA_\mu^a \, d\bar{\Psi} \, d\Psi \, dh \, \exp\left[i \int d^4x \, \mathcal{L} \right] . \tag{2.23}$$

We know that the change in baryon number between the initial time 0 and final time t_f can be written as

$$B(t_f) - B(0) = \int_0^{t_f} dt \int d^3x \langle \partial_\mu j_B^\mu \rangle \tag{2.24}$$

Substituting Eq.(2.21) into the above expression gives

$$B(t_f) - B(0) = \int_0^{t_f} dt \int d^3x \left[\frac{1}{2} 48 \delta(0) \int dA_\mu^a d\bar{\Psi} d\Psi dh \exp\left(i \int d^4x \mathcal{L} \right) \right] \tag{2.25}$$

Performing the space-time integration

$$\int_0^{t_f} dt \int d^3x = Vt_f$$

and defining the effective action as:

$$\int dA_\mu^a d\bar{\Psi} d\Psi dh \exp\left(i \int d^4x \mathcal{L}\right) = \exp[i\Gamma(T)],$$

we obtain

$$B(t_f) - B(0) \approx 24Vt_f\delta(0)\exp[i\Gamma(T)] \tag{2.26}$$

The baryon asymmetry of the Universe is defined as[39]

$$\eta = \frac{n_B}{n_\gamma}, \tag{2.27}$$

where

$$n_B = n_b - \bar{n}_b \tag{2.28}$$

is the baryon number density and n_γ is the photon number density.
The baryon number density can be written as

$$n_B = \frac{B}{V}, \tag{2.29}$$

where B is the total baryon number and V is the spatial volume. Thus the baryon asymmetry becomes

$$\eta = \frac{B/V}{n_\gamma}. \tag{2.30}$$

Now consider the net baryon production

$$\Delta_B B$$

which represents the baryon asymmetry generated in the process. Therefore the baryon density can be written as[39]

$$n_B = \frac{\Delta_B B}{V}. \tag{2.31}$$

Substituting this into the definition of η gives

$$\eta = \frac{\Delta_B B}{V n_\gamma}. \tag{2.32}$$

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In the early Universe one may approximate

$$n_\gamma \sim \frac{1}{V}, \tag{2.33}$$

so that the volume factors approximately cancel. Hence

$$\eta \approx \Delta_B B, \tag{2.34}$$

where

$$\Delta_B B = \frac{1}{2}\left(\Delta B - \Delta \bar{B}\right) \tag{2.35}$$

represents the CP–violating baryon asymmetry.
From eqn (2.24) we have

$$\Delta B = 24 V t_f \delta(0) e^{i\Gamma(T)}. \tag{2.36}$$

Similarly for antibaryons

$$\Delta \bar{B} = 24 V t_f \delta(0) e^{i\bar{\Gamma}(T)}. \tag{2.37}$$

Using eqn.(2.33)
and substituting the expressions above gives

$$\Delta_B B = \frac{1}{2}\left[24 V t_f \delta(0) e^{i\Gamma(T)} - 24 V t_f \delta(0) e^{i\bar{\Gamma}(T)}\right]. \tag{2.38}$$

Factoring out the common terms we obtain

$$\Delta_B B = 12 V t_f \delta(0)\left[e^{i\Gamma(T)} - e^{i\bar{\Gamma}(T)}\right], \tag{2.39}$$

Instead of using the baryon-to-photon ratio

$$\eta = \frac{n_B}{n_\gamma},$$

cosmologists often uses the baryon-to-entropy ratio

$$\eta_1 = \frac{n_B}{s}, \tag{2.40}$$

where s is the entropy density.
In the radiation "dominated era", the entropy density is approximately

$$s \approx g_* n_\gamma, \tag{2.41}$$

where

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$$g_* = 106.75 \tag{2.42}$$

is the number of relativistic degrees of freedom in the Standard Model.
Therefore

$$\eta_1 = \frac{n_B}{g_* n_\gamma}. \tag{2.43}$$

Using eqn.(2.32),we finally obtain

$$\boxed{\eta_1 \approx \frac{\Delta_B B}{g_*}} \tag{2.44}$$

2.4 Asymmetry of the Exponential of the Effective Action

The effective action of the Standard Model at finite temperature can be obtained from the partition function expressed in terms of the classical Higgs field [7, 36].

In quantum field theory, the partition function (or generating functional) is defined as [34]

$$Z = \int \mathcal{D}\Phi \, e^{iS[\Phi]}, \tag{2.45}$$

where the action of the theory is given by

$$S = \int d^4x \, \mathcal{L}(\Phi), \tag{2.46}$$

and Φ collectively denotes all fields of the theory.
In the Standard Model the fields appearing in the path integral include

- gauge fields A^a_μ,
- fermionic fields Ψ and $\bar{\Psi}$,
- the scalar Higgs field h.

Therefore the functional integration measure becomes

$$\mathcal{D}\Phi = dA^a_\mu \, dh \, d\bar{\Psi} \, d\Psi. \tag{2.47}$$

The effective action Γ is defined through the relation

$$Z = \exp(i\Gamma). \tag{2.48}$$

Thus the exponential of the effective action can be written as

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$$\exp(i\Gamma) = \int dA^a_\mu \, dh \, d\bar{\Psi} \, d\Psi \exp(iS). \tag{2.49}$$

we obtain

$$\exp(i\Gamma) = \int dA^a_\mu \, dh \, d\bar{\Psi} \, d\Psi \exp\left[i \int d^4x \, \mathcal{L}(\Phi) \right]. \tag{2.50}$$

We are interested in baryon asymmetry generated through CP violation [38, 18]. We focus only on the quark contribution to the Lagrangian. It is convenient to remove the CP–violating phase from the quark mixing term and transfer it into the down–quark mass matrix. Let U denote the CKM matrix [8] and d represent the down–type quark fields. Let's change the variable

$$\mathrm{U}d = d', \tag{2.51}$$

The Yukawa Mass Term is defined through

$$\mathcal{L}_{\text{Yukawa}} = \bar{d}_L \, M_d \, d_R + \text{h.c.} \tag{2.52}$$

where M_d is a complex 3×3 matrix in flavour space, generally not diagonal. Now ,to diagonalize it,

$$M_d = V_L \, M_d \, V_R^\dagger \tag{2.53}$$

M_d: diagonal, real, positive entries (m_d, m_s, m_b). V_L, V_R: unitary.

CKM matrix:

$$U = V_{uL}^\dagger \, V_{dL} \tag{2.54}$$

therefore,

$$Ud = d' \quad \Longrightarrow \quad d = U^\dagger d', \qquad \bar{d} = \bar{d}' \, U \tag{2.55}$$

Now,

$$\mathcal{L}_{\text{Yukawa}} = \bar{d} \, M_d \, d \tag{2.56}$$

Substituting $d = U^\dagger d'$ and $\bar{d} = \bar{d}' \, U$:

$$\mathcal{L}_{\text{Yukawa}} = (\bar{d}' \, U) \, M_d \, (U^\dagger d')$$

$$\boxed{\mathcal{L}_{\text{mass}} = \bar{d}' \, U M_d U^\dagger \, d'} \tag{2.57}$$

For convenience we define [22]

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$$A + iB = \mathrm{U} M_d \mathrm{U}^\dagger, \tag{2.58}$$

where A and B denote respectively the real and imaginary parts of the matrix $\mathrm{U} M_d \mathrm{U}^\dagger$. Writing the CKM matrix as

$$\mathrm{U} = \mathrm{U}_1 + i\mathrm{U}_2,$$

with U_1 and U_2 representing the real and imaginary parts of U, we obtain

$$A + iB = (\mathrm{U}_1 M_d \mathrm{U}_1^\dagger + \mathrm{U}_2 M_d \mathrm{U}_2^\dagger) + i(\mathrm{U}_2 M_d \mathrm{U}_1^\dagger - \mathrm{U}_1 M_d \mathrm{U}_2^\dagger).$$

Using this notation, the relevant part of the Lagrangian can be written as

$$\mathcal{L} = \mathcal{L}_1 - \bar{d}\,(A + iB)\,d \tag{2.59}$$

$$\mathcal{L} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion kinetic}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}} \tag{2.60}$$

where

- $\mathcal{L}$ denotes the full Lagrangian density of the theory.
- $\mathcal{L}_1$ represents the remaining part of the Standard Model Lagrangian that does not contain the CP–violating contribution from the down–quark mass term.

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Using the decomposition of the Lagrangian

$$\mathcal{L} = \mathcal{L}_1 - \bar{d}(A + iB)d, \tag{2.61}$$

the action becomes

$$\int d^4x\, \mathcal{L} = \int d^4x\, \mathcal{L}_1 - \int d^4x\, \bar{d}Ad - i\int d^4x\, \bar{d}Bd.$$

Substituting this into the path integral (2.48) :

$$e^{i\Gamma} = \int dA_\mu^a\, dh\, d\bar{\Psi}\, d\Psi \exp\left[i\int d^4x(\mathcal{L}_1 - \bar{d}Ad - i\bar{d}Bd)\right].$$

Separating the exponential terms we obtain

$$e^{i\Gamma} = \int dA_\mu^a\, dh\, d\bar{\Psi}\, d\Psi \exp\left[i\int d^4x(\mathcal{L}_1 - \bar{d}Ad)\right] \exp\left[\int d^4x\, \bar{d}Bd\right].$$

The second exponential can be expanded using the series[7, 36]

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

which yields

$$\exp\left[\int d^4x\, \bar{d}Bd\right] = \sum_{n=0}^{\infty} \frac{1}{n!}\left[\int d^4x\, \bar{d}Bd\right]^n.$$

Substituting this expansion into the functional integral gives

$$e^{i\Gamma} = \int dA^a_\mu\, dh\, d\bar{\Psi}\, d\Psi \sum_{n=0}^{\infty} \frac{1}{n!}\left[\int d^4x\, \bar{d}Bd\right]^n \exp\left[i\int d^4x(\mathcal{L}_1 - \bar{d}Ad)\right].$$

Introducing matrix indices, the bilinear term becomes

$$\bar{d}Bd = \bar{d}_i B_{ij} d_j,$$

so that

$$\int d^4x\, \bar{d}Bd = \int d^4x\, \bar{d}_i B_{ij} d_j.$$

Inside the path integral the fermion bilinear can be expressed as a derivative with respect to the mass matrix A. Using

$$\frac{\partial}{\partial A_{ij}} \exp\left[i\int d^4x(-\bar{d}Ad)\right] = i\int d^4x\, \bar{d}_j d_i \exp(\ldots),$$

we obtain the correspondence

$$\bar{d}_i d_j \longrightarrow i\,\frac{\partial}{\partial A_{ij}}.$$

Thus the insertion

$$\bar{d}Bd = B_{ij}\bar{d}_i d_j$$

can be written as

$$iB_{ij}\frac{\partial}{\partial A_{ij}}.$$

Therefore

$$\left[\int d^4x\, \bar{d}Bd\right]^n \to \left[iB_{ij}\frac{\partial}{\partial A_{ij}}\right]^n.$$

Substituting this into the expansion gives

$$e^{i\Gamma} = \int dA^a_\mu\, dh\, d\bar{\Psi}\, d\Psi \sum_{n=0}^{\infty} \frac{1}{n!}\left[iB_{ij}\frac{\partial}{\partial A_{ij}}\right]^n \exp\left[i\int d^4x(\mathcal{L}_1 - \bar{d}Ad)\right].$$

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We now define the effective action corresponding to the mass matrix A

$$e^{i\Gamma(A)} = \int dA^a_\mu \, dh \, d\bar{\Psi} \, d\Psi \exp\left[i \int d^4x (\mathcal{L}_1 - \bar{d}Ad)\right].$$

Thus

$$e^{i\Gamma} = \sum_{n=0}^{\infty} \frac{1}{n!} \left[iB_{ij} \frac{\partial}{\partial A_{ij}}\right]^n e^{i\Gamma(A)}.$$

Next the matrix A is diagonalized using a unitary transformation

$$VAV^\dagger = A_d,$$

where A_d is the diagonal matrix of eigenvalues m'_{dk} and V is unitary matrix. Under this transformation the derivatives satisfy

$$\frac{\partial}{\partial A_{ij}} = V_{ki} V_{jk} \frac{\partial}{\partial m'_{dk}}.$$

Substituting this relation yields

$$e^{i\Gamma} = \sum_{n=0}^{\infty} \left[\frac{i^n}{n!} \left(V_{ki} B_{ij} V^\dagger_{kj} \frac{\partial}{\partial m'_{dk}}\right)^n\right] e^{i\Gamma(A)},$$

The quantity $\Gamma(A)$ denotes the effective action evaluated with the CP–even mass matrix A for the down quarks.[7, 36]

Under the CP transformation the imaginary part of the mass matrix changes sign, while the real part remains unchanged:

$$\mathcal{L}(\mathrm{CP}) = \mathcal{L}_1 - \bar{d}(A - iB)d. \tag{2.62}$$

and so

$$e^{i\bar{\Gamma}} = \int dA^a_\mu \, dh \, d\bar{\Psi} \, d\Psi \exp\left[i \int d^4x (\mathcal{L}_1 - \bar{d}Ad)\right] \exp\left[-\int d^4x \, \bar{d}Bd\right].$$

as before ,

$$\left[\int d^4x \, \bar{d}Bd\right]^n \to \left[-iB_{ij} \frac{\partial}{\partial A_{ij}}\right]^n.$$

Therefore, we can write the expression of CP transformed effective action as:

$$\exp[i\bar{\Gamma}] = \sum_{n=0}^{\infty} \left[\frac{(-i)^n}{n!} \left(V_{ki} B_{ij} V^\dagger_{kj} \frac{\partial}{\partial m'_{d_k}}\right)^n\right] \exp[i\Gamma(A)]. \tag{2.63}$$

We can now compute the difference between the effective action and its CP–conjugate

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[24]

$$\exp[i\Gamma] - \exp[i\bar{\Gamma}] = \sum_{n=0}^{\infty} \left[\frac{i^n - (-i)^n}{n!} \left(V_{ki} B_{ij} V_{kj}^{\dagger} \frac{\partial}{\partial m'_{d_k}} \right)^n \right] \exp[i\Gamma(A)]. \tag{2.64}$$

If n is even, then this vanishes,therefore only odd powers contribute. So writing $n = 2k + 1$, the above expression reduces to [24]

$$\exp[i\Gamma] - \exp[i\bar{\Gamma}] = \sum_{n=2k+1} \left[2\,\frac{i^n}{n!} \left(V_{ki} B_{ij} V_{kj}^{\dagger} \frac{d}{dm'_{d_k}} \right)^n \right] \exp[i\Gamma(A)]. \tag{2.65}$$

2.5 Calculation Of Baryon Asymmetry In The Standard Model

The effective action for a scalar background field Φ is defined from the path integral as

$$e^{i\Gamma[\Phi]} = \int \mathcal{D}\phi \; e^{iS[\phi]}. \tag{2.66}$$

For a constant background field the effective action can be written in terms of the effective potential,

$$\Gamma[\Phi] = -\int d^4x \, V_{\text{eff}}(\Phi, T) \tag{2.67}$$

At one–loop order and finite temperature the effective potential can be decomposed as[24]

$$V_{\text{eff}}(\Phi, T) = V_0(\Phi) + V_1(\Phi) + V_1^T(\Phi, T) \tag{2.68}$$

The Higgs potential of the Standard Model is

$$V_0(\Phi) = -\frac{m^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4 \tag{2.69}$$

Zero temperature one–loop correction

Quantum fluctuations of particles coupled to the Higgs field generate the Coleman–Weinberg correction[24]

$$V_1(\Phi) = \sum_i n_i (-1)^{2s_i} \frac{1}{64\pi^2} m_i^4(\Phi) \left[\ln\left(\frac{m_i^2(\Phi)}{\mu^2} \right) - C_i \right] \tag{2.70}$$

Where i represents the particle, s_i its spin, n_i the number of degrees of freedom

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associated to the specific particle, μ is the renormalization scale and C_i are constants depending on the renormalization scheme. For example $n_g = 3$ for the gauge bosons, $n_q = 12$ for the quarks and $n_l = 4$ for the leptons. For our calculations we will take $C_q = \frac{3}{2}$.

Thermal corrections

At finite temperature the effective potential receives additional contributions. Using the imaginary time formalism and performing the Matsubara sums one obtains [24]

$$V_1^T(\Phi, T) = \sum_{i=\text{boson}} n_i \frac{T^4}{2\pi^2} J_b\left(\frac{m_i^2}{T^2}\right) - \sum_{j=\text{fermion}} n_j \frac{T^4}{2\pi^2} J_f\left(\frac{m_j^2}{T^2}\right). \tag{2.71}$$

The thermal loop functions are defined as

$$J_b(y^2) = \int_0^\infty dx\, x^2 \ln\left[1 + e^{-\sqrt{x^2+y^2}}\right], \tag{2.72}$$

$$J_f(y^2) = \int_0^\infty dx\, x^2 \ln\left[1 - e^{-\sqrt{x^2+y^2}}\right]. \tag{2.73}$$

These functions originate from the Bose–Einstein and Fermi–Dirac distributions for bosonic and fermionic degrees of freedom in the thermal bath.

For a constant classical background field Φ the effective action can be written as

$$\Gamma[\Phi] = \int d^4x\, \mathcal{L}_{\text{eff}}. \tag{2.74}$$

The effective potential is defined as

$$V_{\text{eff}}(\Phi) = -\mathcal{L}_{\text{eff}}. \tag{2.75}$$

Therefore the effective action becomes

$$\Gamma[\Phi] = -\int d^4x\, V_{\text{eff}}(\Phi). \tag{2.76}$$

The spacetime volume integral can be separated as

$$\int d^4x = \int dt \int d^3x = V t_f \tag{2.77}$$

So, the effective action becomes

$$\Gamma(\Phi) = -V\, t_f\, V_{\text{eff}}(\Phi). \tag{2.78}$$

At finite temperature the system is described by a thermal bath. In statistical mechanics the thermal properties of a system are encoded in the partition function

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$$Z = \mathrm{Tr}\left(e^{-\beta H}\right),$$

where H is the Hamiltonian and

$$\beta = \frac{1}{T}.$$

In quantum field theory the time evolution operator has the form

$$e^{-iHt}.$$

In order to relate the quantum evolution operator to the thermal Boltzmann factor $e^{-\beta H}$, one performs a Wick rotation by introducing an imaginary time variable τ such that

$$t \to -i\tau.$$

Under this transformation the evolution operator becomes

$$e^{-iH(-i\tau)} = e^{-\tau H},$$

which has the same structure as the Boltzmann factor appearing in the thermal partition function.[24]

Therefore the Euclidean time interval becomes

$$t_f \sim \frac{1}{T}. \tag{2.79}$$

In thermal equilibrium the correlation length of the system is typically of order

$$\ell \sim \frac{1}{T}. \tag{2.80}$$

Therefore the spatial volume scale behaves as

$$V \sim \left(\frac{1}{T}\right)^3. \tag{2.81}$$

The spacetime volume is therefore

$$Vt_f = \left(\frac{1}{T}\right)^3 \left(\frac{1}{T}\right). \tag{2.82}$$

Thus

$$Vt_f = \frac{1}{T^4}. \tag{2.83}$$

Substituting this result into the expression for the effective action gives[24]

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$$\Gamma(\Phi) = -Vt_f V_{\text{eff}}. \tag{2.84}$$

Therefore

$$\Gamma(\Phi) = -\frac{1}{T^4} V_{\text{eff}}, \tag{2.85}$$

We know from the previous derivations that

$$e^{i\Gamma} = \sum_{n=0}^{\infty} \left[\frac{i^n}{n!} \left(V_{ki} B_{ij} V_{kj}^{\dagger} \frac{\partial}{\partial m'_{dk}} \right)^n \right] e^{i\Gamma(A)}, \tag{2.86}$$

Assuming $\Gamma_{eff}(A) = \frac{-V_{eff}(A)}{T^4}$:

$$\frac{\partial}{\partial m'_{dk}} \exp\left[i\Gamma_{eff}(A)\right] = \frac{-i}{T^4} \frac{\partial V_{eff}}{\partial m'_{dk}} \exp\left[i\Gamma_{eff}(A)\right] \tag{2.87}$$

Applying the derivative operator n times:

$$\left(\frac{\partial}{\partial m'_{dk}} \right)^n \exp\left[i\Gamma_{eff}(A)\right] = (-i/T^4)^n \left(\frac{\partial V_{eff}}{\partial m'_{dk}} \right)^n \exp\left[i\Gamma_{eff}(A)\right] \tag{2.88}$$

From Eq. (2.66) for quarks ($n_q = 12$, $s_i = 1/2$, $C_q = 3/2$): [24]

$$V_1(m'_{dk}) = -\frac{3}{16\pi^2} m'^4_{dk} \left[\ln\left[\frac{m'^2_{dk}}{\mu^2} \right] - \frac{3}{2} \right] \tag{2.89}$$

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Differentiating with respect to m'_{dk}:

$$\frac{\partial V_1}{\partial m'_{dk}} = -\frac{3}{16\pi^2} \left[4m'^3_{dk} \left(\ln\left[\frac{m'^2_{dk}}{\mu^2} \right] - \frac{3}{2} \right) + m'^4_{dk} \left(\frac{2}{m'_{dk}} \right) \right] \tag{2.90}$$

$$\frac{\partial V_1}{\partial m'_{dk}} = -\frac{3}{16\pi^2} \left[4m'^3_{dk} \ln\left[\frac{m'^2_{dk}}{\mu^2} \right] - 4m'^3_{dk} \right] \tag{2.91}$$

$$\frac{\partial V_1}{\partial m'_{dk}} = -\frac{3}{4\pi^2} m'^3_{dk} \left[\ln\left[\frac{m'^2_{dk}}{\mu^2} \right] - 1 \right] \tag{2.92}$$

From Eq(2.71):

$$V_1^T(m'_{dk}) = -\frac{6T^4}{\pi^2} J_f \left(\frac{m'^2_{dk}}{T^2} \right) \tag{2.93}$$

Using the approximation for $y \ll 1$:

$$J_f\left(\frac{m'^2_{dk}}{T^2}\right) \approx \frac{7\pi^4}{360} - \frac{\pi^2}{24}\frac{m'^2_{dk}}{T^2} \tag{2.94}$$

Substituting back into V_1^T:

$$V_1^T(m'_{dk}) \approx -\frac{7\pi^2 T^4}{60} + \frac{1}{4}T^2 m'^2_{dk} \tag{2.95}$$

Differentiating with respect to m'_{dk}:

$$\frac{\partial V_1^T}{\partial m'_{dk}} = \frac{1}{2}m'_{dk}T^2 \tag{2.96}$$

Combining the total derivative of the effective potential (2.92 and 2.88) :

$$\frac{\partial V_{eff}}{\partial m'_{dk}} = -\frac{3}{4\pi^2}m'^3_{dk}\left[\ln\left[\frac{m'^2_{dk}}{\mu^2}\right] - 1\right] + \frac{1}{2}m'_{dk}T^2 \tag{2.97}$$

Substituting into the operator expansion in eqn.(2.82): [24]

$$e^{i\Gamma} = \sum_{n=0}^{\infty}\frac{i^n}{n!}(-i/T4)^n\left[V_{ki}B_{ij}V^t_{kj}\left(-\frac{3}{4\pi^2}m'^3_{dk}\left[\ln\left[\frac{m'^2_{dk}}{\mu^2}\right] - 1\right] + \frac{1}{2}m'_{dk}T^2\right)\right]^n \exp\left[i\Gamma_{eff}(A)\right] \tag{2.98}$$

We now proceed to evaluate the CP–asymmetric contribution using the result obtained in Eq. (2.59), the operator expansion acting on the effective action reads[24]

$$e^{i\bar{\Gamma}} = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\left[V_{ki}B_{ij}V^{\dagger}_{kj}\frac{\partial}{\partial m'_{dk}}\right]^n \exp[i\Gamma_{\text{eff}}(A)]\,.$$

(2.99)

Using the equation (2.93), we have

$$\frac{\partial V_{eff}}{\partial m'_{dk}} = -\frac{3}{4\pi^2}m'^3_{dk}\left[\ln\left[\frac{m'^2_{dk}}{\mu^2}\right] - 1\right] + \frac{1}{2}m'_{dk}T^2 \tag{2.100}$$

Therefore the operator in Eq. (2.100) becomes[24]

$$e^{i\bar{\Gamma}} = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}(-i/T^4)^n\left[V_{ki}B_{ij}V^{\dagger}_{kj}\left(-\frac{3}{4\pi^2}m'^3_{dk}\left(\ln\frac{m'^2_{dk}}{\mu^2} - 1\right) + \frac{1}{2}m'_{dk}T^2\right)\right]^n \exp[i\Gamma_{\text{eff}}(A)].$$

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(2.101)

Now using equation (2.61),we obtain :

$$\exp[i\Gamma] - \exp[i\bar{\Gamma}] = 2 \sum_{n=2k+1} \frac{(i/T^4)^n}{n!} \left[V_{ki} B_{ij} V_{kj}^{\dagger} \left(-\frac{3}{4\pi^2} m_{dk}'^3 \left(\ln \frac{m_{dk}'^2}{\mu^2} - 1 \right) + \frac{1}{2} m_{dk}' T^2 \right) \right]^n \exp[i\Gamma_{\text{eff}}(A)$$

(2.102)

Since the odd series corresponds to the sine expansion

$$\sum_{n=2k+1} \frac{i^n}{n!} x^n = i \sin x,$$

the above expression reduces to

$$\exp[-\Gamma] - \exp[-\bar{\Gamma}] = 2i \sin \left[\frac{V_{ki} B_{ij} V_{kj}}{T^4} \left(-\frac{3}{4\pi^2} m_{dk}'^3 \left(\ln \frac{m_{dk}'^2}{\mu^2} - 1 \right) + \frac{1}{2} m_{dk}' T^2 \right) \right] \exp[-\Gamma_{\text{eff}}(A)].$$

(2.103)

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Finally substituting this result into the baryon asymmetry definition

$$\Delta_B B = 12\, \delta(0)\, V t_f \left[\exp[-\Gamma(T)] - \exp[-\bar{\Gamma}(T)] \right],$$

we obtain

$$\Delta_B B = i\, 24\, \delta(0)\, V t_f \sin \left[\frac{V_{ki} B_{ij} V_{kj}^{\dagger}}{T^4} \left(-\frac{3}{4\pi^2} m_{dk}'^3 \left(\ln \frac{m_{dk}'^2}{\mu^2} - 1 \right) + \frac{1}{2} m_{dk}' T^2 \right) \right] \exp[-\Gamma_{\text{eff}}(A)].$$

(2.104)

Using the relation derived earlier

$$Vt_f = \frac{1}{T^4},$$

the final expression becomes

$$\Delta_B B = i\,24\,\delta(0)\,\frac{1}{T^4}\sin\left[\frac{V_{ki}B_{ij}V_{kj}^\dagger}{T^4}\left(-\frac{3}{4\pi^2}m_{dk}'^3\left(\ln\frac{m_{dk}'^2}{\mu^2}-1\right)+\frac{1}{2}m_{dk}'T^2\right)\right]\exp[-\Gamma_{\text{eff}}(A)].$$

(2.105)

2.6 Numerical Calculation Of The Baryon Asymmetry

In the present analysis all particle masses of the Standard Model are taken at the electroweak scale. The temperature relevant for electroweak baryogenesis is typically [24]

$$T \simeq 100\,\text{GeV}.$$

Furthermore, perturbation theory is most reliable when the renormalization scale is chosen as [24]

$$\mu = \max[m_i(\Phi)].$$

For the Standard Model this corresponds to the top quark mass,

$$\mu = m_t = 173.3\,\text{GeV}.$$

An important exception concerns the down quark masses entering the effective action. These are extracted from the matrix

$$A + iB = UM_dU^\dagger,$$

where U is the CKM matrix and

$$M_d = \begin{pmatrix} m_d & 0 & 0 \\ 0 & m_s & 0 \\ 0 & 0 & m_b \end{pmatrix}$$

is the diagonal mass matrix of the down quarks in the standard model computed at the minimum of the thermal potential.

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The CKM matrix can be decomposed into real and imaginary parts,[24]

$$\mathrm{U} = \mathrm{U}_1 + i\mathrm{U}_2.$$

Substituting this decomposition into the expression for the mass matrix gives[24]

$$\mathrm{U}M_d\mathrm{U}^\dagger = (\mathrm{U}_1 + i\mathrm{U}_2)M_d(\mathrm{U}_1^\dagger - i\mathrm{U}_2^\dagger).$$

Expanding the product yields

$$\mathrm{U}M_d\mathrm{U}^\dagger = \mathrm{U}_1M_d\mathrm{U}_1^\dagger + \mathrm{U}_2M_d\mathrm{U}_2^\dagger + i(\mathrm{U}_2M_d\mathrm{U}_1^\dagger - \mathrm{U}_1M_d\mathrm{U}_2^\dagger).$$

Thus the real and imaginary parts are

$$A = \mathrm{U}_1M_d\mathrm{U}_1^\dagger + \mathrm{U}_2M_d\mathrm{U}_2^\dagger$$

$$B = \mathrm{U}_2M_d\mathrm{U}_1^\dagger - \mathrm{U}_1M_d\mathrm{U}_2^\dagger.$$

The real matrix A can be diagonalized through an orthogonal transformation

$$VAV^T = A_d,$$

where

$$A_d = \begin{pmatrix} m'_{d1} & 0 & 0 \\ 0 & m'_{d2} & 0 \\ 0 & 0 & m'_{d3} \end{pmatrix}.$$

These eigenvalues correspond to the effective down–quark masses appearing in the effective action.[24]

Rescaling of the quark masses

The effective potential is evaluated at the minimum of the thermal Higgs potential. The quark masses therefore scale with the Higgs expectation value,

$$m'_{dk} = z\, m_{dk},$$

where

$$z = \frac{\Phi_{\min}}{v}.$$

Here $v = 246\,\mathrm{GeV}$ is the Higgs vacuum expectation value and $\Phi_{\min}$ denotes the minimum of the temperature–dependent effective potential.

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Using the numerical value

$$\Phi_{\text{min}} = 208.77\,\text{GeV},$$

one obtains

$$z = \frac{208.77}{246} \approx 0.848.$$

Therefore the effective masses become [24]

$$m'_{d1} = z\,m_d = 0.848 \times 4.7 \times 10^{-3} \approx 3.985 \times 10^{-3}\,\text{GeV},$$

$$m'_{d2} = z\,m_s = 0.848 \times 0.096 \approx 0.0814\text{GeV},$$

$$m'_{d3} = z\,m_b = 0.848 \times 4.18 \approx 3.5446\text{GeV}.$$

Thus the eigenvalues entering the effective action are

$$m'_{d1} \simeq 3.985 \times 10^{-3}\,\text{GeV},$$

$$m'_{d2} \simeq 0.0814\,\text{GeV},$$

$$m'_{d3} \simeq 3.5446\,\text{GeV}.$$

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Regularization of the $\delta(0)$ factor

The Dirac delta function in 4D momentum space is defined as:

$$\delta^{(4)}(x-y) = \int \frac{d^4k}{(2\pi)^4} \exp[ik\cdot(x-y)] \tag{2.106}$$

To make the integral finite, we apply a momentum-dependent exponential damping factor (regulator). [24]

Inserting the regulator $\exp\left[\frac{-[i\partial_\mu]^2}{\mu^2}\right]$ and taking the limit $x \to y$:

$$\delta(0) = \lim_{x\to y} \delta(x-y) = \lim_{x\to y} \int \frac{d^4k}{(2\pi)^4} \exp\left[-\frac{(i\partial_\mu)^2}{M^2}\right] \exp\left[ik\cdot(x-y)\right].$$

so that

$$\delta(0)_{\text{reg}} = \int \frac{d^4k}{(2\pi)^4} \exp\left[\frac{-k^2}{\mu^2}\right] \tag{2.107}$$

Here M denotes the ultraviolet cutoff scale introduced through the regulator e^{-k^2/M^2}, while μ represents the renormalization scale. In this work, both scales are effectively identified, i.e., $M \simeq \mu$.

We will use Wick rotation to solve the integral:[24]

Let $k^0 = ik_E^0$. The integration measure transforms as:

$$d^4k = id^4k_E \tag{2.108}$$

The integral becomes purely Euclidean:

$$i \int \frac{d^4k_E}{(2\pi)^4} \exp\left[\frac{-k_E^2}{\mu^2}\right] \tag{2.109}$$

The 4D Euclidean integral separates into four identical 1D Gaussian integrals over each momentum component k_j (where $j = 1, 2, 3, 4$):

$$i \left(\int_{-\infty}^{\infty} \frac{dk_j}{2\pi} \exp\left[\frac{-k_j^2}{\mu^2}\right] \right)^4 \tag{2.110}$$

Using the standard Gaussian integral identity $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ with $a = 1/\mu^2$:

$$\delta(0)_{\text{reg}} = i \left(\frac{\mu}{2\sqrt{\pi}} \right)^4 = i \frac{\mu^4}{16\pi^2} \tag{2.111}$$

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Substituting $\mu = m_t = 172.69$ GeV:

$$\frac{\delta(0)_{\text{reg}}}{T^4} = i \frac{\mu^4}{16\pi^2 T^4}$$

$$\frac{\delta(0)_{\text{reg}}}{T^4} = i \frac{(172.69)^4}{16\pi^2 (100)^4}$$

$$\frac{\delta(0)_{\text{reg}}}{T^4} = i \frac{889683674.5}{157.913 \times 10^8}$$

$$\frac{\delta(0)_{\text{reg}}}{T^4} \approx 0.0563\, i$$

Now, using g* = 106.75 , the relativistic degrees of freedom;

$$\frac{\delta(0)_{\text{reg}}}{g^* T^4} = \frac{1}{106.75}\left(\frac{\delta(0)_{\text{reg}}}{T^4}\right)$$

$$\frac{\delta(0)_{\text{reg}}}{g^* T^4} = \frac{0.056318\, i}{106.75}$$

$$\frac{\delta(0)_{\text{reg}}}{g^* T^4} \approx 0.0005276\, i$$

$$\text{Finally ,we have then : } 24i \times \left(\frac{\delta(0)_{\text{reg}}}{T^4}\right)$$

$$= 24i \times (0.00052757\, i)$$

$$= 0.0126617\, i^2$$

$$= -0.0126617$$

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Evaluation of the effective potential and baryon asymmetry

The finite–temperature effective potential of the Standard Model can be written, retaining the dominant terms relevant for the electroweak phase transition [36]

$$V(\Phi, T) = D(T^2 - T_0^2)\Phi^2 - ET\Phi^3 + \frac{\lambda(T)}{4}\Phi^4. \tag{2.112}$$

where, as before, D, T_0 and E are temperature-independent coefficients, and λ is a slowly varying temperature-dependent function.[36]

Coefficient D

$$D = \frac{2m_W^2 + m_Z^2 + 2m_t^2}{8v^2}. \tag{2.113}$$

Here $v = 246\,\text{GeV}$ is the Higgs vacuum expectation value.

Coefficient E The cubic term arises from the bosonic thermal loops,

$$E = \frac{2m_W^3 + m_Z^3}{4\pi v^3}. \tag{2.114}$$

This term is responsible for generating the first–order electroweak phase transition.

Coefficient B The constant B arises from the zero temperature one–loop corrections to the Higgs potential,[36]

$$B = \frac{3}{64\pi^2 v^4}\left(2m_W^4 + m_Z^4 - 4m_t^4\right). \tag{2.115}$$

Critical temperature parameter The parameter T_0^2 appearing in the quadratic term is

$$T_0^2 = \frac{m_h^2 - 8Bv^2}{4D}. \tag{2.116}$$

Temperature dependent quartic coupling The quartic coupling also receives thermal corrections,

$$\lambda(T) = \lambda - \frac{3}{16\pi^2 v^2}\left[2m_W^4 \ln\left(\frac{m_W^2}{A_B T^2}\right) + m_Z^4 \ln\left(\frac{m_Z^2}{A_B T^2}\right) - 4m_t^4 \ln\left(\frac{m_t^2}{A_F T^2}\right)\right]. \tag{2.117}$$

The constants A_B and A_F originate from the high temperature expansion of the thermal integrals,

$$\ln A_B = \ln a_b - \frac{3}{2}, \qquad \ln A_F = \ln a_f - \frac{3}{2}.$$

Their numerical values are

$$\ln a_b = 5.4076, \qquad \ln a_f = 2.6351.$$

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The Effective Potential and Minimization Condition

The high-temperature expansion of the one-loop effective potential for the Standard Model is given by [36]

$$V(\Phi_c, T) = D(T^2 - T_0^2)\Phi_c^2 - ET\Phi_c^3 + \frac{\lambda(T)}{4}\Phi_c^4 \tag{2.118}$$

To determine the vacuum expectation value at the broken phase minimum (Φ_{min}), we require $\frac{\partial V}{\partial \Phi_c} = 0$:

$$2D(T^2 - T_0^2)\Phi_c - 3ET\Phi_c^2 + \lambda(T)\Phi_c^3 = 0 \tag{2.119}$$

Factoring out the trivial symmetric solution ($\Phi_c = 0$), the broken phase condition is:

$$\lambda(T)\Phi_c^2 - 3ET\Phi_c + 2D(T^2 - T_0^2) = 0 \tag{2.120}$$

Solving this quadratic equation yields the analytical expression for the thermal vacuum

expectation value:

$$\Phi_{\min} = \frac{3ET + \sqrt{9E^2T^2 - 8\lambda(T)D(T^2 - T_0^2)}}{2\lambda(T)} \tag{2.121}$$

We will use following standard electroweak parameters are evaluated at the electroweak scale $T = 100$ GeV:

$$\begin{aligned} v &= 246.22 \text{ GeV} \\ m_W &= 80.385 \text{ GeV} \\ m_Z &= 91.1876 \text{ GeV} \\ m_t &= 173.3 \text{ GeV} \\ m_h &= 125.09 \text{ GeV} \end{aligned}$$

Using the established parameters, we calculate the potential constants:[36]

$$D = \frac{2m_W^2 + m_Z^2 + 2m_t^2}{8v^2} = \frac{2(80.385)^2 + (91.1876)^2 + 2(173.3)^2}{8(246.22)^2} \approx 0.1676 \tag{2.122}$$

$$E = \frac{2m_W^3 + m_Z^3}{4\pi v^3} = \frac{2(80.385)^3 + (91.1876)^3}{4\pi(246.22)^3} \approx 0.00958 \tag{2.123}$$

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$$\begin{aligned} B &= \frac{3}{64\pi^2 v^4}(2m_W^4 + m_Z^4 - 4m_t^4) \\ &= \frac{3}{64\pi^2(246.22)^4}\left[2(80.385)^4 + (91.1876)^4 - 4(173.3)^4\right] \approx -0.00446 \end{aligned} \tag{2.124}$$

$$T_0^2 = \frac{m_h^2 - 8Bv^2}{4D} = \frac{(125.09)^2 - 8(-0.00446)(246.22)^2}{4(0.1676)} \approx 26564 \text{ GeV}^2 \tag{2.125}$$

Thermal Quartic Coupling $\lambda(T)$

We require the logarithmic integration constants:

$$\begin{aligned} \ln(A_B) &= \ln(5.4076) - 1.5 = 3.9076 \\ \ln(A_F) &= \ln(2.6351) - 1.5 = 1.1351 \end{aligned}$$

The tree-level quartic coupling is fixed by the physical Higgs mass:

$$\lambda = \frac{m_h^2}{2v^2} = \frac{(125.09)^2}{2(246.22)^2} = 0.12905 \tag{2.126}$$

The thermal coupling incorporates one-loop high-temperature corrections:

$$\lambda(T) = \lambda - \frac{3}{16\pi^2 v^4}\left[2m_W^4 \ln\left(\frac{m_W^2}{A_B T^2}\right) + m_Z^4 \ln\left(\frac{m_Z^2}{A_B T^2}\right) - 4m_t^4 \ln\left(\frac{m_t^2}{A_F T^2}\right)\right] \tag{2.127}$$

Evaluating the bracketed term explicitly at $T = 100$ GeV:

$$\begin{aligned} B &= 2(80.385)^4 \ln\left(\frac{(80.385)^2}{e^{3.9076}(100)^2}\right) + (91.1876)^4 \ln\left(\frac{(91.1876)^2}{e^{3.9076}(100)^2}\right) \\ &\quad - 4(173.3)^4 \ln\left(\frac{(173.3)^2}{e^{1.1351}(100)^2}\right) \\ &\approx -3.628 \times 10^8 - 2.829 \times 10^8 + 1.299 \times 10^8 \approx -5.158 \times 10^8 \text{ GeV}^4 \end{aligned}$$

Applying the prefactor:

$$\lambda(T) \approx 0.12905 - \left(5.168 \times 10^{-12}\right)\left(-5.158 \times 10^8\right) \approx 0.1317 \tag{2.128}$$

Substituting the derived parameters into Equation 2.121, where the temperature scale difference is $T^2 - T_0^2 = 10000 - 26564 = -16564\ \text{GeV}^2$:

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$$\begin{aligned} \Phi_{\text{min}} &= \frac{3(0.00958)(100) + \sqrt{9(0.00958)^2(100)^2 - 8(0.1317)(0.1676)(-16564)}}{2(0.1317)} \\ &= \frac{2.874 + \sqrt{8.26 + 2925.9}}{0.2634} \approx 216.56 \text{ GeV} \end{aligned} \tag{2.129}$$

2.6.1 Numerical Calculation Manually

We evaluate the total effective potential $V_{\text{eff}}(T, \bar{\Phi})$ at the field minimum $\bar{\Phi}_{\text{min}} = 216.56$ GeV and temperature $T = 100$ GeV. The calculation proceeds in three parts: the tree-level contribution V_0, the one-loop Coleman–Weinberg correction V_1, and the leading finite-temperature correction V_1^T.

Parameters for Tree-level potential:

The quartic coupling and mass parameter are fixed by the physical Higgs and electroweak VEV:

$$\lambda = \frac{m_h^2}{2v^2} = \frac{(125.25\ \text{GeV})^2}{2\,(246.22\ \text{GeV})^2} \approx 0.1294, \tag{2.130}$$

$$m^2 = \lambda v^2 = 0.1294 \times (246.22\ \text{GeV})^2 \approx 7844\ \text{GeV}^2. \tag{2.131}$$

Tree-Level Potential

The Mexican-hat tree-level potential evaluated at Φ_{min} gives

$$\begin{aligned} V_0(\Phi_{\text{min}}) &= -\frac{m^2}{2}\,\Phi_{\text{min}}^2 + \frac{\lambda}{4}\,\Phi_{\text{min}}^4 \\ &= -\frac{7844}{2}(216.56)^2 + \frac{0.1294}{4}(216.56)^4 \\ &\approx -1.840 \times 10^8 + 7.115 \times 10^7 \\ &\approx -1.128 \times 10^8\ \text{GeV}^4. \end{aligned} \tag{2.132}$$

Coleman–Weinberg One-Loop Correction

For the down-type quarks, with effective mass $m_{dk} \approx 4.28$ GeV and renormalization scale $\mu = 173.3$ GeV, the Coleman–Weinberg contribution is

$$\begin{aligned} V_1(\bar{\Phi}_{\text{min}}) &= -\frac{3}{16\pi^2}\,m_{dk}^4\left[\ln\left(\frac{m_{dk}}{\mu}\right) - \frac{3}{2}\right] \\ &= -\frac{3}{16\pi^2}\,(4.28)^4\left[\ln\left(\frac{4.28}{173.3}\right) - \frac{3}{2}\right] \\ &\approx -\frac{3}{16\pi^2} \times 335.5 \times (-5.201) \\ &\approx +33.15\ \text{GeV}^4. \end{aligned} \tag{2.133}$$

This is many orders of magnitude smaller than V_0 and V_1^T, so it contributes negligibly to the total.

Calculation of V_1^T

The corrected one-loop thermal potential, with J_b and J_f properly labeled following the standard convention is:[36]

$$V_1^T = \frac{T^4}{2\pi^2}\left[\sum_{i=\text{boson}} n_i\,J_b\left(\frac{m_i^2}{T^2}\right) - \sum_{j=\text{fermion}} n_j\,J_f\left(\frac{m_j^2}{T^2}\right)\right], \tag{2.134}$$

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with the standard thermal loop functions defined as:

$$J_b(y^2) = \int_0^\infty dx\, x^2 \ln\left[1 - e^{-\sqrt{x^2+y^2}}\right] < 0, \tag{2.135}$$

$$J_f(y^2) = \int_0^\infty dx\, x^2 \ln\left[1 + e^{-\sqrt{x^2+y^2}}\right] > 0. \tag{2.136}$$

Retaining only the first two terms of each high-T expansion $y = \frac{m}{T} \ll 1$:

$$J_b(y^2) \approx -\frac{\pi^4}{45} + \frac{\pi^2}{12}\, y^2, \tag{2.137}$$

$$J_f(y^2) \approx +\frac{7\pi^4}{360} - \frac{\pi^2}{24}\, y^2. \tag{2.138}$$

Substituting $y_i^2 = m_i^2/T^2$ into the truncated expansions:

$$V_1^T = \frac{T^4}{2\pi^2} \sum_{\text{boson}} n_i \left(-\frac{\pi^4}{45} + \frac{\pi^2 m_i^2}{12\, T^2}\right) - \frac{T^4}{2\pi^2} \sum_{\text{fermion}} n_j \left(\frac{7\pi^4}{360} - \frac{\pi^2 m_j^2}{24\, T^2}\right). \tag{2.139}$$

Expanding and collecting terms:

$$V_1^T = \underbrace{-\frac{\pi^2 T^4}{90} N_b}_{\text{Term I}} + \underbrace{\frac{T^2}{24} M_b^2}_{\text{Term II}} - \underbrace{\frac{7\pi^2 T^4}{720} N_f}_{\text{Term III}} + \underbrace{\frac{T^2}{48} M_f^2}_{\text{Term IV}}, \tag{2.140}$$

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where we define the weighted sums: [24]

$$N_b = \sum_{\text{boson}} n_i, \qquad M_b^2 = \sum_{\text{boson}} n_i\, m_i^2, \qquad N_f = \sum_{\text{fermion}} n_j, \qquad M_f^2 = \sum_{\text{fermion}} n_j\, m_j^2. \tag{2.141}$$

Standard Model Particle Content

Bosons (physical degrees of freedom in broken phase):

Particle	n_i	m_i (GeV)	$n_i m_i^2$ (GeV^2)
$W^\pm$	6	80.377	38,763.1
Z^0	3	91.188	24,945.6
γ	2	0	0
Gluons	16	0	0
h	1	125.25	15,687.6
Total	**28**		**79,396.3**

$$N_b = 28, \qquad M_b^2 = 79{,}396 \text{ GeV}^2. \tag{2.142}$$

Fermions (all SM Dirac fermions, 3 generations):

Particle	n_j	m_j (GeV)	$n_j m_j^2$ (GeV^2)
t (3 colors $\times$ 4 spin-charge)	12	172.69	357,861.6
b	12	4.18	209.6
c	12	1.27	19.4
τ	4	1.777	12.6
$u, d, s, e, \mu, \nu_{e,\mu,\tau}$	—	≈ 0	≈ 0
Quarks subtotal	72		
Leptons subtotal	18		
Total	**90**		**358,103**

$$N_f = 90, \qquad M_f^2 = 358{,}103 \text{ GeV}^2. \tag{2.143}$$

Since the masses of $u, d, s, e, \mu, \nu_{e,\mu,\tau}$ are tiny in GeV units ,we ignore thier contributions. As a consistency check, the effective relativistic degrees of freedom:[24]

$$g_* = N_b + \frac{7}{8} N_f = 28 + \frac{7}{8} \times 90 = 28 + 78.75 = 106.75, \quad \checkmark \tag{2.144}$$

which matches the Standard Model value at $T \gtrsim m_t$.

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Numerical Evaluation at $T = 100$ GeV :

At $T = 100$ GeV: $\quad T^2 = 10^4$ GeV^2, $\quad T^4 = 10^8$ GeV^4.

$$-\frac{\pi^2}{90} N_b T^4 = -\frac{9.8696 \times 28}{90} \times 10^8 = -3.0704 \times 10^8 \text{ GeV}^4. \tag{2.145}$$

$$\frac{M_b^2 T^2}{24} = \frac{79{,}396}{24} \times 10^4 = 3308.2 \times 10^4 = +3.308 \times 10^7 \text{ GeV}^4. \tag{2.146}$$

$$-\frac{7\pi^2}{720} N_f T^4 = -\frac{7 \times 9.8696 \times 90}{720} \times 10^8 = -\frac{6217.8}{720} \times 10^8 = -8.636 \times 10^8 \text{ GeV}^4. \tag{2.147}$$

$$\frac{M_f^2 T^2}{48} = \frac{358{,}103}{48} \times 10^4 = 7460.5 \times 10^4 = +7.461 \times 10^7 \text{ GeV}^4. \tag{2.148}$$

Final Sum

Converting all terms to units of 10^8 GeV^4:

$$\begin{aligned} V_1^T &= (-3.0704 + 0.3308 - 8.636 + 0.7461) \times 10^8 \text{ GeV}^4 \\ &= -10.630 \times 10^8 \text{ GeV}^4. \end{aligned} \tag{2.149}$$

$$\boxed{V_1^T(T = 100 \text{ GeV}) \;\approx\; -1.063 \times 10^9 \text{ GeV}^4.} \tag{2.150}$$

Total Effective Potential and Γ_{eff}

Summing all three contributions:[24]

$$\begin{aligned} V_{\text{eff}}(T, \bar{\Phi}_{\text{min}}) &= V_0 + V_1 + V_1^T \\ &\approx \left(-1.128 \times 10^8\right) + \left(33.15\right) + \left(-1.063 \times 10^9\right) \text{ GeV}^4 \\ &\approx (-1.128 - 10.63) \times 10^8 \text{ GeV}^4 \\ &\approx -11.758 \times 10^8 \text{ GeV}^4 \\ &\approx -1.176 \times 10^9 \text{ GeV}^4. \end{aligned} \tag{2.151}$$

The dimensionless ratio that enters the baryon asymmetry computation is

$$\begin{aligned} \Gamma_{\text{eff}}(A) &\equiv -\frac{V_{\text{eff}}}{T^4} = -\frac{-1.176 \times 10^9}{(100)^4} \\ &= \frac{1.176 \times 10^9}{10^8} \approx 11.76, \end{aligned} \tag{2.152}$$

$$e^{-\Gamma_{\text{eff}}(A)} \approx e^{-11.76} \approx 7.8 \times 10^{-6}. \tag{2.153}$$

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Numerical Evaluation of the Sine Term

Starting from Eq. (2.109)

$$\Delta_B B = i\,24\,\delta(0)\,\frac{1}{T^4}\sin\left[\frac{\mathrm{U}_{ki}\,B_{ij}\,\mathrm{U}^\dagger_{kj}}{T^4}\left(-\frac{3}{4\pi^2}\,m'^3_{dk}\left(\ln\frac{m'^2_{dk}}{\mu^2} - 1\right) + \frac{1}{2}\,m'_{dk}\,T^2\right)\right]\exp\left[-\Gamma_{\text{eff}}(A)\right]. \tag{2.154}$$

where

$$B = \mathrm{U}_2 M_d \mathrm{U}_1^\dagger - \mathrm{U}_1 M_d \mathrm{U}_2^\dagger.$$

the baryon number variation takes the form[24]

$$\Delta_B B \;=\; i\,24\,\delta(0)\,\frac{1}{T^4}\;\sin\Big[\mathcal{M}_{ki}\cdot f(m'_{dk})\Big]\;e^{-\Gamma_{\text{eff}}}, \tag{2.155}$$

where the CP-violating matrix element is $\mathcal{M}_{ki} \equiv V_{ki} B_{ij} V^{\dagger}_{kj}$, and the scalar function $f(m'_{dk})$ is defined as

$$f(m'_{dk}) \equiv -\frac{3}{4\pi^2} m'^3_{dk} \left(\ln \frac{m'^2_{dk}}{\mu^2} - 1 \right) + \frac{1}{2} m'_{dk} T^2. \tag{2.156}$$

With $m'_{dk} = 4.28\,\text{GeV}$, $\mu = 173.3\,\text{GeV}$, and $T = 100\,\text{GeV}$, the two terms are evaluated separately.

$$\ln \frac{m'^2_{dk}}{\mu^2} = 2 \ln \frac{4.28}{173.3} = -7.402, \tag{2.157}$$

$$-\frac{3}{4\pi^2} (4.28)^3 (-7.402 - 1) = -\frac{3}{4\pi^2} \times 78.40 \times (-8.402) \approx +50.08\,\text{GeV}^3. \tag{2.158}$$

$$\frac{1}{2} m'_{dk} T^2 = \frac{1}{2} \times 4.28 \times (100)^2 = 21400\,\text{GeV}^3. \tag{2.159}$$

Therefore,we arrive at the result:

$$\boxed{f(m'_{dk}) \approx 50.08 + 21400 \approx 2.145 \times 10^4\,\text{GeV}^3.} \tag{2.160}$$

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Input Parameters

We use the standard Wolfenstein parameterization of the CKM matrix with

$$\lambda = 0.22501, \qquad A = 0.826, \qquad \bar{\rho} = 0.159, \qquad \bar{\eta} = 0.348. \tag{2.161}$$

To leading order, the barred and unbarred parameters are related by

$$\rho = \frac{\bar{\rho}}{1 - \lambda^2/2}, \qquad \eta = \frac{\bar{\eta}}{1 - \lambda^2/2}. \tag{2.162}$$

With $\lambda = 0.22501$, we get

$$1 - \frac{\lambda^2}{2} = 1 - \frac{(0.22501)^2}{2} \simeq 0.97469, \tag{2.163}$$

so

$$\rho \simeq \frac{0.159}{0.97469} \simeq 0.163, \qquad \eta \simeq \frac{0.348}{0.97469} \simeq 0.357. \tag{2.164}$$

We also use

$$m_d = 0.0047\,\text{GeV}, \qquad m_b = 4.18\,\text{GeV}, \qquad T = 100\,\text{GeV}. \tag{2.165}$$

CKM Matrix to $\mathcal{O}(\lambda^3)$

Up to $\mathcal{O}(\lambda^3)$, the CKM matrix is

$$\mathrm{U} \simeq \begin{pmatrix} 1-\frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1-\frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1-\rho-i\eta) & -A\lambda^2 & 1 \end{pmatrix}. \tag{2.166}$$

Write

$$\mathrm{U} = \mathrm{U}_1 + i\mathrm{U}_2, \tag{2.167}$$

where U_1 and U_2 are real matrices. Then

$$\mathrm{U}_1 = \begin{pmatrix} 1-\frac{\lambda^2}{2} & \lambda & A\lambda^3\rho \\ -\lambda & 1-\frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1-\rho) & -A\lambda^2 & 1 \end{pmatrix}, \qquad \mathrm{U}_2 = \begin{pmatrix} 0 & 0 & -A\lambda^3\eta \\ 0 & 0 & 0 \\ -A\lambda^3\eta & 0 & 0 \end{pmatrix}. \tag{2.168}$$

Derivation of the B Matrix

Let

$$M_d = \mathrm{diag}(m_d, m_s, m_b). \tag{2.169}$$

Then

$$\begin{aligned} \mathrm{U}M_d\mathrm{U}^\dagger &= (\mathrm{U}_1 + i\mathrm{U}_2)M_d(\mathrm{U}_1^T - i\mathrm{U}_2^T) \\ &= \left(\mathrm{U}_1 M_d \mathrm{U}_1^T + \mathrm{U}_2 M_d \mathrm{U}_2^T\right) + i\left(\mathrm{U}_2 M_d \mathrm{U}_1^T - \mathrm{U}_1 M_d \mathrm{U}_2^T\right). \end{aligned} \tag{2.170}$$

Therefore the imaginary part is

$$B = \mathrm{U}_2 M_d \mathrm{U}_1^T - \mathrm{U}_1 M_d \mathrm{U}_2^T. \tag{2.171}$$

Since U_2 is already $\mathcal{O}(\lambda^3)$, we keep only the leading terms in λ.
First,

$$\mathrm{U}_2 M_d = \begin{pmatrix} 0 & 0 & -A\lambda^3\eta\, m_b \\ 0 & 0 & 0 \\ -A\lambda^3\eta\, m_d & 0 & 0 \end{pmatrix}. \tag{2.172}$$

Using

$$\mathrm{U}_1^T \simeq \begin{pmatrix} 1-\frac{\lambda^2}{2} & -\lambda & A\lambda^3(1-\rho) \\ \lambda & 1-\frac{\lambda^2}{2} & -A\lambda^2 \\ A\lambda^3\rho & A\lambda^2 & 1 \end{pmatrix}, \tag{2.173}$$

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we obtain, to leading order,

$$X \equiv \mathrm{U}_2 M_d \mathrm{U}_1^T \simeq \begin{pmatrix} 0 & -A^2\lambda^5\eta\, m_b & -A\lambda^3\eta\, m_b \\ 0 & 0 & 0 \\ -A\lambda^3\eta\, m_d & A\lambda^4\eta\, m_d & 0 \end{pmatrix}. \tag{2.174}$$

Because

$$B = X - X^T, \tag{2.175}$$

we get

$$B \simeq \begin{pmatrix} 0 & -A^2\lambda^5\eta\, m_b & -A\lambda^3\eta\,(m_b - m_d) \\ A^2\lambda^5\eta\, m_b & 0 & -A\lambda^4\eta\, m_d \\ A\lambda^3\eta\,(m_b - m_d) & A\lambda^4\eta\, m_d & 0 \end{pmatrix}. \tag{2.176}$$

Numerical Evaluation of B

We now evaluate the powers of λ:

$$\lambda^2 = (0.22501)^2 = 0.0506295, \quad \lambda^3 = (0.22501)^3 = 0.0113921,$$

$$\lambda^4 = (0.22501)^4 = 0.00256335, \quad \lambda^5 = (0.22501)^5 = 0.000576779.$$

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Using

$$A = 0.826, \quad \eta = 0.357, \quad m_d = 0.0047\,\text{GeV}, \quad m_b = 4.18\,\text{GeV},$$

we obtain

$$-A^2\lambda^5\eta\, m_b = -(0.826)^2(0.000576779)(0.357)(4.18) \simeq -5.87 \times 10^{-4}\,\text{GeV},$$

$$-A\lambda^3\eta\,(m_b - m_d) = -(0.826)(0.0113921)(0.357)(4.18 - 0.0047) \simeq -1.40 \times 10^{-2}\,\text{GeV},$$

$$-A\lambda^4\eta\, m_d = -(0.826)(0.00256335)(0.357)(0.0047) \simeq -3.55 \times 10^{-6}\,\text{GeV}.$$

Thus,

$$B \simeq \begin{pmatrix} 0 & -5.87 \times 10^{-4} & -1.40 \times 10^{-2} \\ 5.87 \times 10^{-4} & 0 & -3.55 \times 10^{-6} \\ 1.40 \times 10^{-2} & 3.55 \times 10^{-6} & 0 \end{pmatrix} \text{GeV}.$$

Dominant Contribution to β_k

We start from

$$\beta_k = (\mathrm{U}B\mathrm{U}^\dagger)_{kk}.$$

Expanding explicitly,

$$\beta_k = \sum_{i,j} \mathrm{U}_{ki}\, B_{ij}\, \mathrm{U}^\dagger_{kj}.$$

Since B is real and antisymmetric,

$$B_{ij} = -B_{ji},$$

the terms appear in pairs. In particular, the contribution from $(i,j) = (1,3)$ and $(3,1)$ gives

$$\mathrm{U}_{k1} B_{13} \mathrm{U}^*_{k3} + \mathrm{U}_{k3} B_{31} \mathrm{U}^*_{k1}.$$

Using $B_{31} = -B_{13}$, this becomes

$$B_{13}\big(\mathrm{U}_{k1}\mathrm{U}^*_{k3} - \mathrm{U}_{k3}\mathrm{U}^*_{k1}\big).$$

Now, for any complex number z,

$$z - z^* = 2i\,\mathrm{Im}(z),$$

hence

$$\mathrm{U}_{k1}\mathrm{U}^*_{k3} - \mathrm{U}_{k3}\mathrm{U}^*_{k1} = 2i\,\mathrm{Im}\big(\mathrm{U}_{k1}\mathrm{U}^*_{k3}\big).$$

Therefore,

$$\beta_k \simeq 2i\, B_{13}\, \mathrm{Im}\big(\mathrm{U}_{k1}\mathrm{U}^*_{k3}\big).$$

Finally, using the values of matrix elements from the B matrix:

$$|B_{13}| \sim 10^{-2}, \quad |B_{12}| \sim 10^{-4}, \quad |B_{23}| \sim 10^{-6},$$

So, the dominant contribution comes from the B_{13} element.
The diagonal components may be written as

$$\beta_k = 2i \sum_{i<j} B_{ij}\, \mathrm{Im}(\mathrm{U}_{ki}\mathrm{U}^*_{kj}). \tag{2.177}$$

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For $k = 1$, the dominant contribution is

$$\begin{aligned}\beta_1 &\simeq 2i\,[B_{13}\,\mathrm{Im}(\mathrm{U}_{11}\mathrm{U}_{13}^*) + B_{23}\,\mathrm{Im}(\mathrm{U}_{12}\mathrm{U}_{13}^*)]\\ &\simeq 2i\,B_{13}\,\mathrm{Im}(\mathrm{U}_{11}\mathrm{U}_{13}^*)\\ &\simeq 2i\left(-1.40\times10^{-2}\right)\left(3.27\times10^{-3}\right)\\ &\simeq -9.16\times10^{-5}\,i\,\mathrm{GeV},\end{aligned} \tag{2.178}$$

where

$$\mathrm{Im}(\mathrm{U}_{11}\mathrm{U}_{13}^*) = \left(1-\frac{\lambda^2}{2}\right)A\lambda^3\eta \simeq (0.97469)(0.826)(0.0113921)(0.357) \simeq 3.27\times10^{-3} \tag{2.179}$$

Now define the phase argument

$$\arg_b = \frac{|\beta_b|}{T^4}\,f(m'_{dk}). \tag{2.180}$$

Since

$$T^4 = (100)^4 = 10^8\,\mathrm{GeV}^4, \tag{2.181}$$

we get

$$\begin{aligned}\arg_b &= \frac{9.16\times10^{-5}}{10^8}\,(2.094\times10^4)\\ &= 9.16\times10^{-13}\times2.094\times10^4\\ &\simeq 1.92\times10^{-8}.\end{aligned} \tag{2.182}$$

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Sine of the Phase

Because $|\arg_b| \ll 1$,

$$\sin(\arg_b) \simeq \arg_b \simeq 1.92\times10^{-8}. \tag{2.183}$$

So

$$\boxed{\sin(\arg_b) \simeq 1.92\times10^{-8}}. \tag{2.184}$$

Final Numerical Value

Combining all previously computed factors, the baryon-to-entropy ratio from Eq. (2.109) is

$$\eta_1 = \frac{\Delta_B B}{g_*} = \underbrace{\frac{24i\,\delta(0)_{\mathrm{reg}}}{g_* T^4}}_{-0.01266} \times \underbrace{\sin(\arg_b)}_{1.92\times10^{-8}} \times \underbrace{\exp[-\Gamma_{\mathrm{eff}}(A)]}_{7.8\times10^{-6}}. \tag{2.185}$$

Substituting the numerical values obtained in Eqs. (2.11), (2.153), and (2.184):

$$\begin{aligned}\eta_1 &= (-0.01266) \times (1.92 \times 10^{-8}) \times (7.8 \times 10^{-6}) \\ &= (-2.43 \times 10^{-10}) \times (7.8 \times 10^{-6}) \\ &\simeq -1.89 \times 10^{-15}. \end{aligned} \tag{2.186}$$

$$\boxed{|\eta_1| \;=\; \frac{n_B}{s} \;\simeq\; 1.89 \times 10^{-15}.} \tag{2.187}$$

The negative sign reflects the convention in which the CP-conjugate effective action $\bar{\Gamma}$ exceeds Γ, and the magnitude of η_1 should be compared with the observed baryon-to-entropy ratio

$$\left.\frac{n_B}{s}\right|_{\text{obs}} \approx 10^{-10}. \tag{2.188}$$

The computed value falls short of the observed asymmetry by approximately four orders of magnitude, confirming the well-known result that CKM-induced CP violation within the Standard Model is insufficient to account for the observed baryon asymmetry of the Universe. This conclusion is consistent with the independent estimate $\delta_{CP} \sim 10^{-19}$ obtained via the Jarlskog determinant in Chapter 4, and points unambiguously to the need for physics beyond the Standard Model.[24]

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2.6.2 Numerical Value Obtained With Python Code

```python


"""
Numerical evaluation of baryon asymmetry Delta_BB from Eq. (2.109):

    Delta_BB = i*24*delta(0) * (1/T^4) * sin[ (V_ki B_ij Vt_kj / T^4) * f(m'_dk) ]
                * exp[-Gamma_eff(A)]

All indices summed; three down-quark flavors k=1,2,3.
"""

import numpy as np

#
# 1.  Physical Parameters
#
T   = 100.0        # GeV  (EW baryogenesis temperature)
```

```
v    = 246.22       # GeV  (Higgs VEV)
mW   = 80.385       # GeV
mZ   = 91.1876      # GeV
mt   = 173.3        # GeV  (also renormalization scale )
mh   = 125.09       # GeV
mu   = mt           # renormalization scale

md_phys = 0.0047  # GeV
ms_phys = 0.096   # GeV
mb_phys = 4.18    # GeV

# Wolfenstein parameters
lam     = 0.22501
A_w     = 0.826
rho_bar = 0.159
eta_bar = 0.348

rho = rho_bar / (1 - lam**2/2)
eta = eta_bar / (1 - lam**2/2)

print("=" * 60)
print("BARYON ASYMMETRY - NUMERICAL EVALUATION")
print("=" * 60)
print(f"\nWolfenstein:  ={lam}, A={A_w}, ={rho:.4f}, ={eta:.4f}")

#
# 2.  CKM Matrix  U = U1 + i U2  (Wolfenstein, O(³))
#
lam2, lam3, lam4 = lam**2, lam**3, lam**4

U = np.array([
    [1 - lam2/2,                  lam,                  A_w*lam3*(rho - 1j*eta)   ],
    [-lam,                         1 - lam2/2,           A_w*lam2                ],
    [A_w*lam3*(1 - rho - 1j*eta),  -A_w*lam2,            1.0 + 0j                ]
], dtype=complex)

U1 = U.real
U2 = U.imag

```

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```python
print("\n CKM real part U1 ")
print(np.array2string(U1, precision=6))
print("\n CKM imag part U2 ")
print(np.array2string(U2, precision=6))

#
# 3.  Effective potential minimum  _min
#
D     = (2*mW**2 + mZ**2 + 2*mt**2) / (8*v**2)
E_c   = (2*mW**3 + mZ**3)           / (4*np.pi*v**3)
B_c   = 3/(64*np.pi**2*v**4) * (2*mW**4 + mZ**4 - 4*mt**4)
T0_sq = (mh**2 - 8*B_c*v**2)        / (4*D)

# thermal quartic coupling
ln_AB = np.log(5.4076) - 1.5   # = 3.9076  (from high-T expansion constants)
ln_AF = np.log(2.6351) - 1.5   # = 1.1351

lam_tree = mh**2 / (2*v**2)
bracket  = (2*mW**4 * np.log(mW**2 / (np.exp(ln_AB)*T**2))
            +  mZ**4 * np.log(mZ**2 / (np.exp(ln_AB)*T**2))
            - 4*mt**4 * np.log(mt**2 / (np.exp(ln_AF)*T**2)))
lam_T    = lam_tree - 3/(16*np.pi**2*v**4) * bracket

disc    = 9*E_c**2*T**2 - 8*lam_T*D*(T**2 - T0_sq)
Phi_min = (3*E_c*T + np.sqrt(disc)) / (2*lam_T)

z = Phi_min / v  # rescaling factor

print(f"\n Effective Potential Minimum ")
print(f"  D      = {D:.6f}")
print(f"  E      = {E_c:.6f}")
print(f"  T0²    = {T0_sq:.2f} GeV²")
print(f"  (T)    = {lam_T:.6f}")
print(f"  _min   = {Phi_min:.4f} GeV")
print(f"  z      = _min/v = {z:.6f}")

#
# 4.  Rescaled down-quark masses
#
Md = np.diag([md_phys, ms_phys, mb_phys])

# physical rescaled masses (for reference)
```

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```python
m_ref = z * np.array([md_phys, ms_phys, mb_phys])
print(f"\n Rescaled Physical Masses  (z × m_phys) ")
for name, m in zip(["d","s","b"], m_ref):
    print(f"  m'_{name} = {m:.5e} GeV")

#
# 5.  A and B matrices
#     A + iB = U Md U†   (real and imaginary parts)
#
A_mat = U1 @ Md @ U1.T + U2 @ Md @ U2.T   # real symmetric
B_mat = U2 @ Md @ U1.T - U1 @ Md @ U2.T   # real antisymmetric

print(f"\n B matrix (GeV) ")
print(np.array2string(B_mat, precision=5))

#
# 6.  Diagonalize A  →  V A V = diag(m'_d1, m'_d2, m'_d3)
#     numpy eigh: columns of evec are eigenvectors  V = evec.T
#
m_dk_prime, evec = np.linalg.eigh(A_mat)   # eigenvalues in ascending order
V = evec.T                                 # V such that V @ A @ V.T = diag

print(f"\n Eigenvalues of A  (effective quark masses, GeV) ")
for k, m in enumerate(m_dk_prime):
    print(f"  m'_d{k+1} = {m:.6e}")

# sanity check: V @ A @ V.T should be diagonal
diag_check = V @ A_mat @ V.T
print(f"\n  Diagonalization check  max|off-diag| = "
      f"{np.max(np.abs(diag_check - np.diag(np.diag(diag_check))))):.2e}  ")

#
# 7.  _k = (V B V)_{kk}   (full matrix contraction, no approximation)
#
VBVt = V @ B_mat @ V.T          # real antisymmetric matrix
beta = np.diag(VBVt)            # diagonal elements (must be 0 for real antisymm)

# _k must vanish for real antisymmetric B and real orthogonal V
# because (VAV)_{kk} = 0 for antisymmetric A.
# The CP-violating structure requires VBV† with complex V from thesis Eq.(2.161)
```

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```
# Here V is real  _k = 0 identically.
# Thesis avoids this by using the full complex V_{ki} = U_{ki}:

print(f"\n _k = (V B V)_kk  [real V, antisymm B → must vanish] ")
for k, b in enumerate(beta):
    print(f"  _{k+1} = {b:.4e}")

#
# 8.  Full contraction  _{k,i,j} U_{ki} B_{ij} U*_{kj}
#     This is what the thesis actually uses (Eq. 2.109 + context)
#     with V = U (the CKM matrix itself, complex)
#
# _k = (U B U†)_{kk}  where U is the full complex CKM matrix
UBUdag = U @ B_mat @ U.conj().T
beta_full = np.diag(UBUdag)

print(f"\n _k = (U B U†)_kk  [complex CKM, full contraction] ")
for k, b in enumerate(beta_full):
    print(f"  _{k+1} = {b:.6e}")

#
# 9.  f(m'_dk) for each eigenvalue
#
def f_func(m, mu, T):
    """Eq. (2.146): f(m') = -3/(4²) m'³ (ln(m'²/²)-1) + ½ m' T²"""
    return (-3/(4*np.pi**2)) * m**3 * (np.log(m**2/mu**2) - 1) + 0.5*m*T**2

# Use eigenvalues of A for m'_dk
f_vals = np.array([f_func(abs(m), mu, T) for m in m_dk_prime])
print(f"\n f(m'_dk)  [GeV³] ")
for k, (m, f) in enumerate(zip(m_dk_prime, f_vals)):
    print(f"  f(m'_d{k+1}) at m'={abs(m):.4e} GeV  →  {f:.5e} GeV³")

#
# 10. Sine argument  =  _k  _k · f(m'_dk) / T
#
T4 = T**4
sine_arg = np.sum(beta_full * f_vals) / T4

print(f"\n Sine Argument ")
print(f"  _k _k f(m'_k) / T = {sine_arg:.6e}  (complex)")
print(f"  |sine_arg|          = {abs(sine_arg):.6e}")
```

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```python

sine_val = np.sin(sine_arg)
print(f"  sin(arg)              = {sine_val:.6e}")

#
# 11. Regularized  (0)
#     (0)_reg = i  / (16²)   [Eq. (2.101)]
#
delta0 = 1j * mu**4 / (16*np.pi**2)
delta0_over_T4 = delta0 / T4

print(f"\n Regularized (0) ")
print(f"  (0)_reg     = {delta0:.5e}")
print(f"  (0)/T       = {delta0_over_T4:.5e}")

#
# 12. Effective potential  Veff  and  _eff
#
# -- V0 --
lam0   = mh**2 / (2*v**2)
m2_par = lam0 * v**2
V0 = -m2_par/2 * Phi_min**2 + lam0/4 * Phi_min**4

# -- V1 Coleman-Weinberg (dominant: top quark) --
def CW_fermion(n, m, mu):
    if m == 0: return 0.0
    return -n/(64*np.pi**2) * m**4 * (np.log(m**2/mu**2) - 1.5)

def CW_boson(n, m, mu):
    if m == 0: return 0.0
    return  n/(64*np.pi**2) * m**4 * (np.log(m**2/mu**2) - 5/6)

V1 = (CW_boson(6, mW, mu) + CW_boson(3, mZ, mu) + CW_boson(1, mh, mu)
    + CW_fermion(12, mt,      mu)
    + CW_fermion(12, mb_phys, mu)
    + CW_fermion(12, 1.27,    mu)
    + CW_fermion(4,  1.777,   mu))

# -- V1^T thermal (high-T expansion, Eq. 2.130) --
Nb  = 28
M2b = 6*mW**2 + 3*mZ**2 + mh**2
Nf  = 90
```

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```
M2f = 12*mt**2 + 12*mb_phys**2 + 12*1.27**2 + 4*1.777**2

VT1 = ( -np.pi**2*T**4/90 * Nb
        + T**2/24 * M2b
        - 7*np.pi**2*T**4/720 * Nf
        + T**2/48 * M2f )

Veff      = V0 + V1 + VT1
Gamma_eff = -Veff / T4
exp_neg_G = np.exp(-Gamma_eff)

print(f"\n Effective Potential ")
print(f"  V0          = {V0:.5e} GeV")
print(f"  V1          = {V1:.5e} GeV")
print(f"  V1^T        = {VT1:.5e} GeV")
print(f"  Veff        = {Veff:.5e} GeV")
print(f"  _eff       = {Gamma_eff:.5f}")
print(f"  exp(-_eff) = {exp_neg_G:.5e}")

#
# 13. Final  B_B  from Eq. (2.109)
#     B_B = i · 24 · (0)/T · sin(arg) · exp(_eff)
#
Delta_BB = 1j * 24 * delta0_over_T4 * sine_val * exp_neg_G

print(f"\n{'=' * 60}")
print(f"FINAL RESULTS")
print(f"{'=' * 60}")
print(f"  B_B (complex)    = {Delta_BB:.5e}")
print(f"  Re(B_B)          = {Delta_BB.real:.5e}")
print(f"  Im(B_B)          = {Delta_BB.imag:.5e}")

g_star = 106.75
eta1 = Delta_BB.real / g_star

print(f"\n   = nB/s = B_B / g* = {eta1:.5e}")
print(f"\n  Observed value:   8.7 × 10¹¹")
print(f"  Ratio (computed / observed): {abs(eta1) / 8.7e-11:.3e}")
print(f"{'=' * 60}")
```

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Numerical Output Of Python Code

```
============================================================
FINAL RESULTS
============================================================
  ΔB_B (complex)    = -1.33725e-31-2.98526e-13j
  Re(ΔB_B)          = -1.33725e-31
  Im(ΔB_B)          = -2.98526e-13

  η₁ = nB/s = ΔB_B / g* = -1.25269e-33

  Observed value: η₁ ≈ 8.7 × 10⁻¹¹
  Ratio (computed / observed): 1.440e-23
============================================================
```

Figure 2.1: Python numerical output for baryon asymmetry calculation.

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Chapter 3

CP Violation And CKM Matrix

In this chapter, we basically discuss C and P symmetries and the CP operations ,CP violations in the case of weak interactions. I have given detailed explanation of Standard Model allowed CP violation through the CKM Matrix calculations using Standard Model Lagrangian.

3.1 CP Symmetry And CP Operations

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CP Symmetry :-

Charge Conjugation C:

1. Flips following:

- $q \to -q$ (electric charge)
- $B \to -B$ (Baryon no.)
- $L \to -L$ (Lepton no.)
- Gauge charges (colour, weak, hyper charges, etc) also get flipped in sign.

2. Mass, momentum ($\vec{p}$), spin (s) stay the same.

Example:

$$C\,|e^-(\vec{p},s)\rangle = |e^+(\vec{p},s)\rangle$$

- It creates a left-handed positron from left-handed electron.
- In the weak int., this RH particle doesn't couple.

Parity P:

1. It flips spatial coordinates: $\vec{x} = -\vec{x}$

- $\vec{p} \to -\vec{p}$
- Spin doesn't change, because spin is an *axial vector*.

$$P\,|\psi(\vec{p}, s)\rangle = |\psi(-\vec{p}, s)\rangle$$

$$P\,|\text{helicity}\rangle \to -|\text{helicity}\rangle$$

(helicity $\equiv s \cdot \hat{p}$)

CP : Operation

When we apply both, we get the full transformation required to turn a physical "matter" state into a physical "antimatter" state in the Weak sector:

$$CP \equiv C \times P$$

$$CP\,|\text{Particle}_{LH}\rangle \longrightarrow |\text{Antiparticle}_{RH}\rangle$$

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 # Neutrino: We observe only LH neutrinos (ν_L) and only RH antineutrinos ($\bar{\nu}_R$).
$P|\nu_L\rangle = |\nu_R\rangle$, Parity flips helicity
But ν_R doesn't interact via the weak force, it is sterile, so P is violated.[24]

$C|\nu_L\rangle = |\bar{\nu}_L\rangle$, charge conjugation flips internal quantum no. like L
But the weak force doesn't couple to $\bar{\nu}_L$, so C is violated.

Now,

$$CP|\nu_L\rangle \longrightarrow |\bar{\nu}_R\rangle$$

Therefore, the CP operation changes observed particle ν_L into observed antiparticle $\bar{\nu}_R$, and so CP is a better symmetry than C or P alone.

CP violation and SM Lagrangian :-

The full SM Lagrangian consists of three parts: [40]

$$\mathcal{L}_{SM} = \mathcal{L}_{Kinetic} + \mathcal{L}_{Higgs} + \mathcal{L}_{Yukawa}$$

where

$$\mathcal{L}_{Kinetic} = i\bar{\psi}(\partial^{\mu}\gamma_{\mu})\psi \quad \rightarrow \text{for spinor field}$$

$$\bar{\psi} = \psi^{\dagger}\gamma^{0}$$

The spinor fields ψ are the three fermion generations, each consisting of the following five representations:

$$Q^{I}_{Li}(3,2,+1/6), \quad u^{I}_{Ri}(3,1,+2/3), \quad d^{I}_{Ri}(3,1,-1/3)$$
$$L^{I}_{i}(1,2,-1/2), \quad L^{I}_{Ri}(1,1,-1)$$

Where $i \rightarrow 1,2,3 = i$ three generations and superscript I implies that the fermion fields are expressed in the interaction basis. So there are $3 \times 5 = 15$ representations but effectively there are only 12 fermion particles in the standard model. This is explained below.[34]

These are actually the **Group Theoretical definitions** of the fundamental particles in the Standard Model.

In SM, we don't define a particle by its name (like "electron"); we define it by **how it interacts** with the three fundamental forces.

The notation format used here is:

$$\text{Field}(\text{Strong}, \text{Weak}, \text{Hypercharge})$$

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Specifically:

$$\Psi(\text{SU}(3)_C, \text{SU}(2)_L, \text{U}(1)_Y)$$

Here is the breakdown of the filed notations of particles:

First Number: $\text{SU}(3)_C$ (Color/Strong Force)

- **3:** It is a **Triplet**. It has "Color" (Red, Green, Blue). These are **Quarks**.
- **1:** It is a **Singlet**. It is "Colorless". These are **Leptons**.

Second Number: $\text{SU}(2)_L$ (Weak Isospin)

- **2:** It is a **Doublet**. It interacts with the $W^{\pm}$ bosons. This creates the "Weak Force."
- **1:** It is a **Singlet**. It **does not** feel the Charged Weak force.

Note: Notice that only the Left-handed (L) fields are Doublets (2), while the Right-handed (R) fields are Singlets (1).

Third Number: $\mathrm{U}(1)_Y$ (Weak Hypercharge)

This is a number related to electric charge (Q) which is related to hypercharge Y as:

$$Q = I_3 + Y$$

(where I_3 is $+1/2$ for the top member of a doublet, and $-1/2$ for the bottom).

Meaning Of Each Symbol

The Quarks families

1. $Q^I_{Li}(3, 2, +1/6)$

- The Left-Handed Quark Doublet.
  - **3:** It's a quark (feels strong force).
  - **2:** It's a doublet (contains both up-type and down-type, e.g., $\begin{pmatrix} u_L \\ d_L \end{pmatrix}$).
  - **+1/6:** Its hypercharge.
- **Example:** For up left-handed quark: u_L ($I_3 = +1/2$), Charge $Q = 1/2 + 1/6 = 4/6 = +2/3$.

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2. $u^I_{Ri}(3, 1, +2/3)$

- The Right-Handed Up-Type Quark.
  - **3:** Still a quark.
  - **1:** Singlet! It travels alone. It cannot emit a W boson (it doesn't interact weakly).
  - **+2/3:** Hypercharge is equal to its electric charge (since $I_3 = 0$ for singlets).

3. $d^I_{Ri}(3, 1, -1/3)$

- The Right-Handed Down-Type Quark.
  - **3:** Quark.
  - **1:** Singlet (No weak interaction).
  - **-1/3:** Matches electric charge.

The Leptons families

4. $L^I_{Li}(1, 2, -1/2)$

- The Left-Handed Lepton Doublet (e.g., $\begin{pmatrix} \nu_e \\ e_L \end{pmatrix}$).
  - **1:** Colorless (Does not feel strong force).
  - **2:** Doublet (Pairs a neutrino with an electron).
  - **-1/2:** Hypercharge.
- **Example:** For electron e_L ($I_3 = -1/2$), Charge $Q = -1/2 + (-1/2) = -1$.

5. $L^I_{Ri}(1, 1, -1)$

- The Right-Handed Charged Lepton (e.g., e_R).
  - **1:** Colorless.
  - **1:** Singlet (Right-handed electrons don't feel the weak force!).
  - **-1:** Electric charge.

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Note : - "Why don't we just group u_L and u_R together and call it an Up Quark?"

In Quantum Field Theory, we can't do that because they have different quantum numbers. Q_L has Weak Isospin $I = 1/2$. u_R has Weak Isospin $I = 0$.

This is exactly why we need the Higgs field. The Higgs is also a Doublet ($I = 1/2$). It acts as the bridge that connects the Left (Doublet) to the Right (Singlet), allowing them to mix and acquire mass without breaking the rules.[39]

In reality, there are 12 femion particles (6 quarks + 6 leptons) instead of 15 :

$$3 \text{ Generations} \times 5 \text{ Representations per generation} = 15 \text{ Representations total.}$$

However, these 15 pieces combine to form fewer distinct particles because some pieces get "glued together".

Let's look at Generation 1 (the fermions inside the 5 representations):[39]

- Q_L (Doublet): Contains the **left** half of the Up quark AND the **left** half of the Down quark.
- u_R (Singlet): Contains the **right** half of the Up quark.

- d_R (Singlet): Contains the **right** half of the Down quark.
- L_L (Doublet): Contains the **left** half of the Electron AND the Neutrino.
- e_R (Singlet): Contains the **right** half of the Electron.

Result:

- The Up halves ($u_L + u_R$) join to make **1 physical Up quark**.
- The Down halves ($d_L + d_R$) join to make **1 physical Down quark**.
- The Electron halves ($e_L + e_R$) join to make **1 physical Electron**.
- The Neutrino (only has a left half!) stays as **1 physical Neutrino**.

The Final Count (Per Generation):

$$\text{Up} + \text{Down} + \text{Electron} + \text{Neutrino} = 4 \text{ Particles.}$$

Total for the SM:

$$3 \text{ Generations} \times 4 \text{ Particles} = \mathbf{12} \text{ Matter Particles.}$$

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The interaction terms are obtained by imposing gauge invariance by replacing the partial derivative by the covariant derivative:

$$\mathcal{L}_{\text{kinetic}} = i\bar{\psi}(D_\mu \gamma^\mu)\psi$$

with the covariant derivative defined as

$$D_\mu = \partial_\mu + ig_s G_a^\mu L_a + igW_b^\mu \sigma_b + ig' B^\mu Y,$$

where L_a are the Gell-Mann matrices and σ_b are the Pauli matrices. G_a^μ, W_b^μ and B^μ are the eight gluon fields, the three weak interaction bosons and the single hypercharge boson, respectively.[39]

3.2 Derivation of Charged Current Interaction For Left-handed Quarks:

Now, let us derive the charged current interaction term explicitly.[40] We start with the kinetic term for the left-handed quark doublet Q_{Li}^I. As discussed in the previous

section, the relevant term in the Standard Model Lagrangian is:

$$\mathcal{L}_{\text{kinetic,weak}}(Q_L) = \mathcal{L}_{\text{kinetic}} = i\bar{\psi}(D_\mu \gamma^\mu)\psi = i\overline{Q^I_{Li}}\gamma_\mu D^\mu Q^I_{Li} \tag{3.1}$$

where the covariant derivative D^μ for the electroweak interaction (focusing on the $\mathrm{SU}(2)_L$ part) is defined as:

$$D^\mu = \partial^\mu + \frac{i}{2} g W^\mu_b \sigma_b \tag{3.2}$$

Expanding the Fields:
To solve this, we first explicitly write out the components of our fields. The quark doublet Q^I_{Li} in the interaction basis is:

$$Q^I_{Li} = \begin{pmatrix} u^I_{Li} \\ d^I_{Li} \end{pmatrix} \tag{3.3}$$

where u and d represent the up-type and down-type quarks.

The term $W^\mu_b \sigma_b$ involves a summation over the index $b = 1, 2, 3$. We can combine the gauge fields W^1, W^2, W^3 with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ into a single 2×2 matrix:

$$W \cdot \sigma = W^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + W^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + W^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.4}$$

Summing these components element-by-element gives us the interaction matrix:

$$W \cdot \sigma = \begin{pmatrix} W^3 & W^1 - iW^2 \\ W^1 + iW^2 & -W^3 \end{pmatrix} \tag{3.5}$$

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Substituting into the Lagrangian:
Now, we substitute this matrix back into the Lagrangian expression (1.1), we can separate the expression into a Kinetic part (involving ∂^μ) and an Interaction part (involving W^μ):

$$\begin{aligned} \mathcal{L} = & \underbrace{i \begin{pmatrix} \bar{u} & \bar{d} \end{pmatrix} \gamma_\mu \partial^\mu \begin{pmatrix} u \\ d \end{pmatrix}}_{\text{Kinetic Term}} \\ & \underbrace{+ i \begin{pmatrix} \bar{u} & \bar{d} \end{pmatrix} \gamma_\mu \left(\frac{i}{2} g \begin{pmatrix} W^3 & W^1 - iW^2 \\ W^1 + iW^2 & -W^3 \end{pmatrix} \right) \begin{pmatrix} u \\ d \end{pmatrix}}_{\text{Interaction Term}} \end{aligned} \tag{3.6}$$

The factors $i \times i$ give -1, so the coefficient simplifies to $-\frac{g}{2}$:

$$\mathcal{L}_{\text{int}} = -\frac{g}{2} \begin{pmatrix} \bar{u} & \bar{d} \end{pmatrix} \gamma_\mu \begin{pmatrix} W^3 & W^1 - iW^2 \\ W^1 + iW^2 & -W^3 \end{pmatrix} \begin{pmatrix} u \\ d \end{pmatrix} \tag{3.7}$$

Matrix Multiplication:
We perform the matrix multiplication in two steps. First, multiplying the matrix by the column vector:[39]

$$\begin{pmatrix} W^3 & W^1 - iW^2 \\ W^1 + iW^2 & -W^3 \end{pmatrix} \begin{pmatrix} u \\ d \end{pmatrix} = \begin{pmatrix} W^3 u + (W^1 - iW^2)d \\ (W^1 + iW^2)u - W^3 d \end{pmatrix}$$

Next, we multiply this result by the row vector $\begin{pmatrix} \bar{u} & \bar{d} \end{pmatrix}$ from the left from equation(4.7):

$$\begin{aligned} &= \bar{u}[W^3 u + (W^1 - iW^2)d] + \bar{d}[(W^1 + iW^2)u - W^3 d] \\ &= \bar{u}W^3 u - \bar{d}W^3 d + \bar{u}(W^1 - iW^2)d + \bar{d}(W^1 + iW^2)u \end{aligned} \tag{3.8}$$

The fields W^1 and W^2 are not the physical particles we observe. We define the physical charged bosons W^+ and W^- as linear combinations of them:

$$W^-_\mu = \frac{1}{\sqrt{2}}(W^1_\mu - iW^2_\mu) \quad \Rightarrow \quad (W^1 - iW^2) = \sqrt{2}W^- \tag{3.9}$$

$$W^+_\mu = \frac{1}{\sqrt{2}}(W^1_\mu + iW^2_\mu) \quad \Rightarrow \quad (W^1 + iW^2) = \sqrt{2}W^+ \tag{3.10}$$

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Finally, we substitute these physical fields back into our interaction expression. The terms involving W^3 correspond to neutral currents (Z boson interactions), while the off-diagonal terms give us the charged currents:[39]

- Term involving $\bar{u}d$:

$$-\frac{g}{2}\gamma_\mu[\bar{u}(W^1 - iW^2)d] = -\frac{g}{2}\gamma_\mu[\bar{u}(\sqrt{2}W^-)d] = -\frac{g}{\sqrt{2}}\bar{u}\gamma_\mu W^{-\mu} d$$

- Term involving $\bar{d}u$:

$$-\frac{g}{2}\gamma_\mu[\bar{d}(W^1 + iW^2)u] = -\frac{g}{2}\gamma_\mu[\bar{d}(\sqrt{2}W^+)u] = -\frac{g}{\sqrt{2}}\bar{d}\gamma_\mu W^{+\mu} u$$

Combining the kinetic parts and these derived interaction terms, we arrive exactly

at the Lagrangian form from equation (1.7) :

$$\mathcal{L} = \underbrace{i\overline{u^I_{iL}}\gamma_\mu\partial^\mu u^I_{iL} + i\overline{d^I_{iL}}\gamma_\mu\partial^\mu d^I_{iL}}_{\text{Kinetic}} \underbrace{-\frac{g}{\sqrt{2}}\overline{u^I_{iL}}\gamma_\mu W^{-\mu} d^I_{iL} - \frac{g}{\sqrt{2}}\overline{d^I_{iL}}\gamma_\mu W^{+\mu} u^I_{iL}}_{\text{Charged Current Interaction}} + \dots \tag{3.11}$$

This confirms the origin of the charged current interactions in the Standard Model.

3.3 The Higgs Mechanism and Yukawa Couplings

To understand how these fermions acquire mass (since explicit mass terms like $m\bar{\psi}\psi$ are forbidden by gauge invariance), we must introduce the Higgs Lagrangian:

$$\mathcal{L}_{\text{Higgs}} = (D_\mu\phi)^\dagger(D^\mu\phi) - \mu^2\phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2 \tag{3.12}$$

where ϕ is an isospin doublet with hypercharge $Y = +1/2$:

$$\phi(x) = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{3.13}$$

The coupling of the Higgs to the gauge fields follows from the covariant derivative in the kinetic term,the fermions acquire mass through the **Yukawa couplings**. These are interaction terms we add "by hand" between the Higgs field and the fermions:[16]

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$$\begin{aligned} -\mathcal{L}_{Yukawa} &= Y_{ij}\overline{\psi_{Li}}\phi\psi_{Rj} + h.c. \\ &= Y^d_{ij}\overline{Q^I_{Li}}\phi d^I_{Rj} + Y^u_{ij}\overline{Q^I_{Li}}\tilde{\phi}u^I_{Rj} + Y^l_{ij}\overline{L^I_{Li}}\phi l^I_{Rj} + h.c. \end{aligned} \tag{3.14}$$

with

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \phi^* = \begin{pmatrix} (\phi^+)^* \\ (\phi^0)^* \end{pmatrix} = \begin{pmatrix} \phi^- \\ \phi^{0*} \end{pmatrix}$$

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_2\phi^* = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} \phi^- \\ \phi^{0*} \end{pmatrix} = \begin{pmatrix} -i\,\phi^{0*} \\ i\,\phi^- \end{pmatrix}$$

$$i\,\sigma_2\phi^* = i\begin{pmatrix} -i\,\phi^{0*} \\ i\,\phi^- \end{pmatrix} = \begin{pmatrix} \phi^{0*} \\ -\phi^- \end{pmatrix}$$

$$\tilde{\phi} \equiv i\sigma_2\phi^* = \begin{pmatrix} \phi^{0*} \\ -\phi^- \end{pmatrix}$$

so we have

$$\tilde{\phi} = i\sigma_2\phi^* = \begin{pmatrix} \overline{\phi^0} \\ -\phi^- \end{pmatrix}.$$

The matrices Y^d_{ij}, Y^u_{ij} and Y^l_{ij} are arbitrary complex matrices that operate in flavour space, giving rise to couplings between different families, or quark mixing, and thus to the field of flavour physics.[16]

We now write the term $Y^d_{ij}\overline{Q^I_{Li}}\phi d^I_{Rj}$ explicitly in the matrix form:

$$Y^d_{ij}\,\overline{Q^I_{Li}}\,\phi\, d^I_{Rj} = Y^d_{ij}\,\overline{(u\ d)^I_{iL}} \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} d^I_{Rj} \tag{3.15}$$

$$= \begin{pmatrix} Y_{11}\,\overline{(u\ d)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} & Y_{12}\,\overline{(u\ d)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} & Y_{13}\,\overline{(u\ d)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \\ Y_{21}\,\overline{(c\ s)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} & Y_{22}\,\overline{(c\ s)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} & Y_{23}\,\overline{(c\ s)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \\ Y_{31}\,\overline{(t\ b)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} & Y_{32}\,\overline{(t\ b)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} & Y_{33}\,\overline{(t\ b)^I_L}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \end{pmatrix} \begin{pmatrix} d^I_R \\ s^I_R \\ b^I_R \end{pmatrix} \tag{3.16}$$

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The indices i,j take values 1,2,3 , so we got 3 by 3 matrix.[16]

3.4 Spontaneous Symmetry Breaking Of Higgs Field and Mass Generation

For a time $t > 10^{-12}$ s after the Big Bang, the Universe has cooled below 10^{16} K and its ground state breaks the $\mathrm{SU}(2) \otimes \mathrm{U}(1)$ symmetry.

From eqn 1.12 , by flipping sign of mass for symmetry breaking, we have :

$$\mathcal{L}_{\text{Higgs}} = (D_\mu\phi)^\dagger(D^\mu\phi) + \mu^2\,\phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2$$

$$V(\phi) = -\mu^2\,\phi^\dagger\phi + \lambda(\phi^\dagger\phi)^2$$

$$\frac{\partial V}{\partial(\phi^\dagger\phi)} = -\mu^2 + 2\lambda(\phi^\dagger\phi) = 0$$

$$\Rightarrow \qquad \phi^\dagger\phi = \frac{\mu^2}{2\lambda}$$

$$\mu^2 > 0, \qquad \lambda > 0$$

$$\langle\phi^\dagger\phi\rangle = \frac{v^2}{2} = \frac{\mu^2}{2\lambda}$$

$$\Rightarrow \qquad v^2 = \frac{\mu^2}{\lambda}$$

$$\phi_0 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}, \qquad v = \sqrt{\frac{\mu^2}{\lambda}}$$

The minimum in the potential of the Higgs field described in eqn. (1.12) is not at $(\phi)_0 = 0$, but at $(\phi)_0 = v = \sqrt{\frac{\mu^2}{\lambda}}$. We'll choose to break the symmetry with a new ground state at

$$(\phi_1)_0^2 = \frac{\mu^2}{\lambda}, \qquad (\phi_2)_0 = 0, \qquad (\phi_3)_0 = 0, \qquad (\phi_4)_0 = 0.$$

(3.17)

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Writing $(\phi_1)_0 = \left(\frac{\mu^2}{\lambda}\right)^{1/2} = v$ we then have a ground state field

$$(\phi)_0 = \begin{pmatrix}\phi^+\\ \phi^0\end{pmatrix}_0 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}.$$

(3.18)

We could search for excitations from this ground state field in a very general way by allowing every component to vary. However, we are at liberty to perform a different gauge transformation at each point in spacetime, and thus reduce the excited field to the much simpler unitary gauge form : [16]

$$\phi(x) = \begin{pmatrix}0\\ \frac{v+h(x)}{\sqrt{2}}\end{pmatrix}.$$

(3.19)

Thus, when the Higgs field undergoes spontaneous symmetry breaking, it acquires a vacuum expectation value (VEV) v. We can choose the unitary gauge where the field is written as:

$$\phi(x) \xrightarrow{\text{sym. breaking}} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{3.20}$$

From eqn.(3.14), the Yukawa Lagrangian for quarks only is :

$$-\mathcal{L}_{\text{Yukawa}}^{\text{quarks}} = Y_{ij}^{d}\, \overline{Q}_{Li}^{I}\, \phi\, d_{Rj}^{I} + Y_{ij}^{u}\, \overline{Q}_{Li}^{I}\, \tilde{\phi}\, u_{Rj}^{I} + \text{h.c.}$$

with

$$Q_{Li}^{I} = \begin{pmatrix} u_{Li}^{I} \\ d_{Li}^{I} \end{pmatrix}, \qquad \phi(x) = \begin{pmatrix} 0 \\ \dfrac{v + h(x)}{\sqrt{2}} \end{pmatrix}, \qquad \tilde{\phi}(x) = \begin{pmatrix} \dfrac{v + h(x)}{\sqrt{2}} \\ 0 \end{pmatrix},$$

now using these equations, we obtain :

$$\overline{Q_{Li}^{I}}\, \phi = (\overline{u_{Li}^{I}}, \overline{d_{Li}^{I}}) \begin{pmatrix} 0 \\ \dfrac{v + h(x)}{\sqrt{2}} \end{pmatrix} = \overline{d_{Li}^{I}}\, \frac{v + h(x)}{\sqrt{2}},$$

$$\overline{Q_{Li}^{I}}\, \tilde{\phi} = (\overline{u_{Li}^{I}}, \overline{d_{Li}^{I}}) \begin{pmatrix} \dfrac{v + h(x)}{\sqrt{2}} \\ 0 \end{pmatrix} = \overline{u_{Li}^{I}}\, \frac{v + h(x)}{\sqrt{2}}.$$

Hence the Yukawa Lagrangian becomes

$$-\mathcal{L}_{\text{Yukawa}}^{\text{quarks}} = Y_{ij}^{d}\, \overline{d_{Li}^{I}}\, \frac{v + h(x)}{\sqrt{2}}\, d_{Rj}^{I} + Y_{ij}^{u}\, \overline{u_{Li}^{I}}\, \frac{v + h(x)}{\sqrt{2}}\, u_{Rj}^{I} + \text{h.c.}$$

Separating the vacuum and fluctuation pieces,

$$-\mathcal{L}_{\text{Yukawa}}^{\text{quarks}} = Y_{ij}^{d}\, \overline{d_{Li}^{I}}\, \frac{v}{\sqrt{2}}\, d_{Rj}^{I} + Y_{ij}^{u}\, \overline{u_{Li}^{I}}\, \frac{v}{\sqrt{2}}\, u_{Rj}^{I} + h.c.$$

$$+ Y_{ij}^{d}\, \overline{d_{Li}^{I}}\, \frac{h(x)}{\sqrt{2}}\, d_{Rj}^{I} + Y_{ij}^{u}\, \overline{u_{Li}^{I}}\, \frac{h(x)}{\sqrt{2}}\, u_{Rj}^{I} + \text{h.c.}$$

The first line yields fermion mass terms, while the second line describes Higgs–fermion interactions. Defining the quark mass matrices

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$$M^d_{ij} \equiv Y^d_{ij}\,\frac{v}{\sqrt{2}}, \qquad M^u_{ij} \equiv Y^u_{ij}\,\frac{v}{\sqrt{2}},$$

we finally obtain

$$-\mathcal{L}^{\text{quarks}}_{\text{mass}} = M^d_{ij}\,\overline{d^I_{Li}}\,d^I_{Rj} + M^u_{ij}\,\overline{u^I_{Li}}\,u^I_{Rj} + \text{h.c.},$$

together with the Higgs–fermion interaction terms:[16]

$$-\mathcal{L}_{h\text{–fermion}} = \frac{h(x)}{v}\left(M^d_{ij}\,\overline{d^I_{Li}}\,d^I_{Rj} + M^u_{ij}\,\overline{u^I_{Li}}\,u^I_{Rj}\right) + \text{h.c.}.$$

The interaction terms of the fermion fields with the Higgs field, $\bar{q}q\,h(x)$, will be omitted in what follows, since we are now interested only in the fermion mass terms. Since M^d and M^u are in general complex, non-diagonal 3×3 matrices in flavour space, the above Lagrangian does not yet describe fermions with definite masses. To obtain proper mass terms, the matrices M^d and M^u must be diagonalized.[16]

Note:-

The notation convention "+ h.c." is written out so that formulas for Lagrangians and Hamiltonians, which conventionally are both required to be Hermitian operators.

The expression

$$\mathcal{L} = A + B + C + \text{h.c.}$$

means

$$\mathcal{L} = A + B + C + A^\dagger + B^\dagger + C^\dagger.$$

This makes the Lagrangian Hermitian.

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Diagonalization of the Mass Matrices

Any complex matrix M can be diagonalized by two unitary matrices V_L and V_R such that

$$M_{\text{diag}} = V_L\,M\,V_R^\dagger$$

Applying this to the quark mass matrices, we introduce unitary matrices V^d_L, V^d_R and V^u_L, V^u_R satisfying

$$M^d_{\text{diag}} = V^d_L\,M^d\,V^{d\dagger}_R, \qquad M^u_{\text{diag}} = V^u_L\,M^u\,V^{u\dagger}_R$$

By definition, the diagonal matrices have the form [39]

$$M^d_{\text{diag}} = \begin{pmatrix} m_d & 0 & 0 \\ 0 & m_s & 0 \\ 0 & 0 & m_b \end{pmatrix}, \qquad M^u_{\text{diag}} = \begin{pmatrix} m_u & 0 & 0 \\ 0 & m_c & 0 \\ 0 & 0 & m_t \end{pmatrix}.$$

Redefinition of quark fields

We now redefine the quark fields by *unitary flavour rotations*:

$$d_{Li} = (V^d_L)_{ij}\, d^I_{Lj}, \qquad d_{Ri} = (V^d_R)_{ij}\, d^I_{Rj},$$

$$u_{Li} = (V^u_L)_{ij}\, u^I_{Lj}, \qquad u_{Ri} = (V^u_R)_{ij}\, u^I_{Rj}.$$

Equivalently, in matrix notation,

$$d_L = V^d_L\, d^I_L, \qquad d_R = V^d_R\, d^I_R, \qquad u_L = V^u_L\, u^I_L, \qquad u_R = V^u_R\, u^I_R.$$

The inverse transformations are

$$d^I_L = V^{d\dagger}_L\, d_L, \qquad d^I_R = V^{d\dagger}_R\, d_R, \qquad u^I_L = V^{u\dagger}_L\, u_L, \qquad u^I_R = V^{u\dagger}_R\, u_R,$$

using the unitarity conditions

$$V^{d\dagger}_L V^d_L = \mathbf{1}, \qquad V^{d\dagger}_R V^d_R = \mathbf{1}, \qquad V^{u\dagger}_L V^u_L = \mathbf{1}, \qquad V^{u\dagger}_R V^u_R = \mathbf{1}.$$

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Diagonalization of the mass Lagrangian

Substituting the inverse transformations into the mass terms, we obtain[39]

$$d^I_{Li} = (V^{d\dagger}_L)_{ij}\, d_{Lj}, \qquad d^I_{Ri} = (V^{d\dagger}_R)_{ij}\, d_{Rj},$$

into the down-type quark mass term, we obtain

$$M^d_{ij}\, \overline{d^I_{Li}}\, d^I_{Rj} = M^d_{ij}\, \overline{\left((V^{d\dagger}_L)_{ik}\, d_{Lk}\right)}\, \left((V^{d\dagger}_R)_{jl}\, d_{Rl}\right).$$

Taking the Dirac adjoint and using

$$\overline{(V^{d\dagger}_L)_{ik}\, d_{Lk}} = (V^{d\dagger}_L)^*_{ik}\, \overline{d_{Lk}} = (V^d_L)_{ki}\, \overline{d_{Lk}},$$

this becomes

$$M^d_{ij}\, \overline{d^I_{Li}}\, d^I_{Rj} = \overline{d_{Lk}}\, (V^d_L)_{ki}\, M^d_{ij}\, (V^{d\dagger}_R)_{jl}\, d_{Rl}.$$

In matrix notation, using

$$d_L^I = V_L^{d\dagger}\, d_L, \qquad d_R^I = V_R^{d\dagger}\, d_R,$$

the mass term can be written as

$$\overline{d_L^I}\, M^d\, d_R^I = \overline{(V_L^{d\dagger} d_L)}\; M^d\; (V_R^{d\dagger} d_R).$$

Taking the Dirac adjoint in matrix form,

$$\overline{(V_L^{d\dagger} d_L)} = \overline{d_L}\, (V_L^{d\dagger})^\dagger = \overline{d_L}\, V_L^d,$$

we obtain

$$\overline{d_L^I}\, M^d\, d_R^I = \overline{d_L}\, V_L^d\, M^d\, V_R^{d\dagger}\, d_R = \overline{d_L}\, \left(V_L^d\, M^d\, V_R^{d\dagger}\right) d_R.$$

Defining the diagonal mass matrix by

$$M^d_{\text{diag}} \equiv V_L^d\, M^d\, V_R^{d\dagger},$$

we finally arrive at

$$\overline{d_L^I}\, M^d\, d_R^I = \overline{d_L}\, M^d_{\text{diag}}\, d_R.$$

$$M^d_{ij}\, \overline{d_{Li}^I}\, d_{Rj}^I = \overline{d_{Lk}}\, (V_L^d)_{ki}\, M^d_{ij}\, (V_R^{d\dagger})_{jl}\, d_{Rl}.$$

In matrix notation this reads

$$\overline{d_L^I}\, M^d\, d_R^I = \overline{d_L}\, \left(V_L^d\, M^d\, V_R^{d\dagger}\right) d_R = \overline{d_L}\, M^d_{\text{diag}}\, d_R.$$

Similarly, for the up-type quarks,

$$\overline{u_L^I}\, M^u\, u_R^I = \overline{u_L}\, \left(V_L^u\, M^u\, V_R^{u\dagger}\right) u_R = \overline{u_L}\, M^u_{\text{diag}}\, u_R.$$

Therefore, the Yukawa Lagrangian becomes

$$-\mathcal{L}^{\text{quarks}}_{\text{Yukawa}} = \overline{d_{Li}^I}\, M^d_{ij}\, d_{Rj}^I + \overline{u_{Li}^I}\, M^u_{ij}\, u_{Rj}^I + \text{h.c.} + \cdots$$

$$= \overline{d_{Li}}\, (V_L^d)_{ik}\, M^d_{kl}\, (V_R^{d\dagger})_{lj}\, d_{Rj} + \overline{u_{Li}}\, (V_L^u)_{ik}\, M^u_{kl}\, (V_R^{u\dagger})_{lj}\, u_{Rj} + \text{h.c.} + \cdots$$

$$= \overline{d_{Li}}\, (M^d_{\text{diag}})_{ij}\, d_{Rj} + \overline{u_{Li}}\, (M^u_{\text{diag}})_{ij}\, u_{Rj} + \text{h.c.} + \cdots .$$

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We now write the weak interaction Lagrangian in terms of the quark mass eigenstates instead of the interaction (weak) eigenstates. Before any field redefinitions, the charged current interaction among left-handed quarks originating from the kinetic term is[39]

$$\mathcal{L}_{\text{kinetic, cc}}(Q_L) = \frac{g}{\sqrt{2}}\,\overline{u^I_{Li}}\,\gamma^\mu W^-_\mu\, d^I_{Li} + \frac{g}{\sqrt{2}}\,\overline{d^I_{Li}}\,\gamma^\mu W^+_\mu\, u^I_{Li} + \cdots .$$

Using the inverse flavour rotations previously introduced,

$$u^I_{Li} = (V_L^{u\dagger})_{ij}\, u_{Lj}, \qquad d^I_{Li} = (V_L^{d\dagger})_{ij}\, d_{Lj},$$

we substitute these into the charged current term:

$$\overline{u^I_{Li}}\,\gamma^\mu d^I_{Li} = \overline{\left((V_L^{u\dagger})_{ij}\, u_{Lj}\right)}\,\gamma^\mu\,\left((V_L^{d\dagger})_{ik}\, d_{Lk}\right).$$

Taking the Dirac adjoint correctly and using

$$\overline{(V_L^{u\dagger})_{ij}\, u_{Lj}} = (V_L^{u\dagger})^*_{ij}\,\overline{u_{Lj}} = (V_L^u)_{ji}\,\overline{u_{Lj}},$$

this becomes

$$\overline{u^I_{Li}}\,\gamma^\mu d^I_{Li} = \overline{u_{Lj}}\,(V_L^u)_{ji}\,\gamma^\mu\,(V_L^{d\dagger})_{ik}\, d_{Lk}.$$

Hence the charged current Lagrangian can be written as

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$$\mathcal{L}_{\text{kinetic, cc}}(Q_L) = \frac{g}{\sqrt{2}}\,\overline{u_{Lj}}\,(V_L^u V_L^{d\dagger})_{jk}\,\gamma^\mu W^-_\mu\, d_{Lk} + \frac{g}{\sqrt{2}}\,\overline{d_{Lj}}\,(V_L^d V_L^{u\dagger})_{jk}\,\gamma^\mu W^+_\mu\, u_{Lk} + \cdots .$$

Definition of the CKM matrix

We now define the unitary 3×3 CKM matrix as:

$$V_{\text{CKM}} \;\equiv\; V_L^u\, V_L^{d\dagger}.$$

With this definition, the charged current interaction becomes

$$\mathcal{L}_{\text{kinetic, cc}}(Q_L) = \frac{g}{\sqrt{2}}\,\overline{u_{Li}}\,(V_{\text{CKM}})_{ij}\,\gamma^\mu W^-_\mu\, d_{Lj} + \frac{g}{\sqrt{2}}\,\overline{d_{Li}}\,(V^\dagger_{\text{CKM}})_{ij}\,\gamma^\mu W^+_\mu\, u_{Lj} + \cdots .$$

We now clarify the physical difference between the charged current interaction written in the weak interaction basis and the same interaction expressed in the quark mass eigenstate basis.

In the weak interaction basis, the charged current interaction is given by[39]

$$\mathcal{L}_{\text{kinetic,cc}}(Q_L) = -\frac{g}{\sqrt{2}}\,\overline{u^I_{Li}}\,\gamma^\mu W^-_\mu\, d^I_{Li} + \text{h.c.}$$

Here the fields u^I_{Li} and d^I_{Li} denote the interaction (weak) eigenstates. Since the same flavour index i appears on both the up-type and down-type fields, the interaction is diagonal in flavour space. Writing this expression explicitly in terms of generations, we obtain

$$-\frac{g}{\sqrt{2}}\left(\overline{u^I_L}\gamma^\mu W^-_\mu d^I_L + \overline{c^I_L}\gamma^\mu W^-_\mu s^I_L + \overline{t^I_L}\gamma^\mu W^-_\mu b^I_L\right) + \text{h.c.}$$

*** Thus, in the weak basis each up-type quark couples only to the down-type quark of the same generation. There are no couplings between different families, and no flavour mixing is present. This form of the interaction follows directly from the $\mathrm{SU}(2)_L$ gauge symmetry and is flavour diagonal.

After diagonalizing the quark mass matrices and expressing the Lagrangian in terms of the mass eigenstates, the charged current interaction becomes

$$\mathcal{L}_{\text{kinetic,cc}}(Q_L) = -\frac{g}{\sqrt{2}}\,\overline{u_{Li}}\,(V_{\text{CKM}})_{ij}\,\gamma^\mu W^-_\mu\, d_{Lj} + \text{h.c.}$$

In this expression the fields u_{Li} and d_{Lj} denote the quark mass eigenstates. The matrix V_{CKM} is, in general, non-diagonal, and therefore connects different flavour indices. Expanding this interaction explicitly, we obtain[39]

$$\begin{aligned}-\frac{g}{\sqrt{2}}\big[&\overline{u_L}(V_{ud}d_L + V_{us}s_L + V_{ub}b_L)\\ &+\overline{c_L}(V_{cd}d_L + V_{cs}s_L + V_{cb}b_L)\\ &+\overline{t_L}(V_{td}d_L + V_{ts}s_L + V_{tb}b_L)\big]\gamma^\mu W^-_\mu + \text{h.c.}\end{aligned}$$

We see that each up-type quark now connects not to just one down-type quark, but to a combination of all down-type quarks. The coefficients of this combination are exactly the elements of the CKM matrix.

All observable phenomena of quark flavor physics, including strangeness production, heavy-flavor decays, meson mixing, and CP violation, come entirely from this single change in the charged current interaction.[39]

Choice of interaction basis

By convention, one chooses the interaction eigenstates to coincide with the mass eigenstates in the up-type sector. This corresponds to setting

$$u^I_i = u_i.$$

With this choice, the down-type interaction eigenstates are related to the mass

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eigenstates by

$$d_i^I = (V_{\text{CKM}})_{ij}\, d_j.$$

Explicit matrix form

Writing this relation explicitly in vector form, we obtain

$$\begin{pmatrix} d^I \\ s^I \\ b^I \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}.$$

This matrix is the Cabibbo–Kobayashi–Maskawa (CKM) mixing matrix. Each element V_{ij} represents the strength with which an up-type quark u_i couples to a down-type quark d_j through the charged weak interaction.[**arlskog1985zpc**]

Thus quark mixing and CP violation originate from the mismatch between the diagonalizations of the up- and down-type quark mass matrices.

3.5 Diagonalization and the CKM Matrix

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The mass matrices M^d and M^u are generally not diagonal. To find the physical particles (mass eigenstates) u, c, t and d, s, b, we must diagonalize these matrices using bi-unitary transformations.

We introduce four unitary matrices $V_L^u, V_R^u, V_L^d, V_R^d$ such that:

$$M^u_{\text{diag}} = V_L^u M^u V_R^{u\dagger}, \qquad M^d_{\text{diag}} = V_L^d M^d V_R^{d\dagger} \tag{3.21}$$

We can now relate the **interaction eigenstates** (q^I) to the physical **mass eigenstates** (q) via these rotation matrices. Based on the requirement that M_{diag} is diagonal, the states are related by:

$$u_{Li} = (V_L^u)_{ij} u_{Lj}^I \implies u_L^I = (V_L^{u\dagger}) u_L \tag{3.22}$$

$$d_{Li} = (V_L^d)_{ij} d_{Lj}^I \implies d_L^I = (V_L^{d\dagger}) d_L \tag{3.23}$$

(And similarly for the right-handed fields).

Derivation of the CKM Matrix:

Now, we must see how this basis change affects the Charged Current interaction we derived in the previous section. Recall the charged current in the interaction basis:

$$\mathcal{L}_{cc} = \frac{g}{\sqrt{2}} \overline{u_{Li}^I} \gamma_\mu W^{+\mu} d_{Li}^I + h.c. \tag{3.24}$$

We substitute the mass eigenstates into this expression. Using $\overline{u_L^I} = \overline{(V_L^{u\dagger} u_L)} = \overline{u_L} V_L^u$, we get:

$$\begin{aligned}\mathcal{L}_{cc} &= \frac{g}{\sqrt{2}} \left(\overline{u_L} V_L^u\right) \gamma_\mu W^{+\mu} \left(V_L^{d\dagger} d_L\right) + h.c. \\ &= \frac{g}{\sqrt{2}} \overline{u_{Li}} \gamma_\mu W^{+\mu} \left[\sum_k (V_L^u)_{ik} (V_L^{d\dagger})_{kj}\right] d_{Lj} + h.c. \end{aligned} \tag{3.25}$$

The term in the brackets is the famous **Cabibbo-Kobayashi-Maskawa (CKM) Matrix**:

$$V_{\text{CKM}} = V_L^u V_L^{d\dagger} \tag{3.26}$$

This matrix is not diagonal (because V^u and V^d are generally different), meaning the weak charged current can change the flavor of quarks (e.g., $s \to u$). The Lagrangian in the mass basis is finally:

$$\mathcal{L}_{cc} = \frac{g}{\sqrt{2}} \overline{u_L} \gamma_\mu W^+ V_{\text{CKM}} d_L + h.c. \tag{3.27}$$

Explicitly, this connects the up-type quarks (u, c, t) to the down-type quarks (d, s, b):

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix} \tag{3.28}$$

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This misalignment between the mass basis and the interaction basis is the sole source of CP violation in the Standard Model (provided the CKM matrix has a non-vanishing complex phase).

3.5.1 CP violation

We now examine once more the Yukawa part of the Lagrangian,[39]

$$-\mathcal{L}_{\text{Yukawa}} = Y_{ij} \overline{\psi_{Li}} \, \phi \, \psi_{Rj} + \text{h.c.} = Y_{ij} \overline{\psi_{Li}} \, \phi \, \psi_{Rj} + Y_{ij}^* \overline{\psi_{Rj}} \, \phi^\dagger \, \psi_{Li}.$$

Under the combined CP transformation, the fermion and scalar fields transform as

$$\psi_{Li}(x) \xrightarrow{CP} (\psi_{Ri})^{CP}(x'), \qquad \psi_{Rj}(x) \xrightarrow{CP} (\psi_{Lj})^{CP}(x'), \qquad \phi(x) \xrightarrow{CP} \phi^\dagger(x'),$$

so that the bilinear transforms according to

$$CP\left(\overline{\psi_{Li}} \, \phi \, \psi_{Rj}\right) = \overline{\psi_{Rj}} \, \phi^\dagger \, \psi_{Li}.$$

Applying the CP transformation to the Yukawa Lagrangian, we obtain

$$CP(-\mathcal{L}_{\text{Yukawa}}) = Y_{ij}\,\overline{\psi_{Rj}}\,\phi^\dagger\,\psi_{Li} + Y^*_{ij}\,\overline{\psi_{Li}}\,\phi\,\psi_{Rj}.$$

The Lagrangian is invariant under CP if and only if this expression coincides with the original one. This requires

$$Y_{ij} = Y^*_{ij},$$

that is, all Yukawa couplings must be real. Therefore, complex Yukawa matrices constitute a necessary source of CP violation in the Standard Model.

We now examine the charged current interaction in the basis of quark mass eigenstates. The charged current part of the kinetic Lagrangian is

$$\mathcal{L}_{\text{kinetic,cc}}(Q_L) = \frac{g}{\sqrt{2}}\,\overline{u_{Li}}\,V_{ij}\,\gamma^\mu W^-_\mu\,d_{Lj} + \frac{g}{\sqrt{2}}\,\overline{d_{Li}}\,V^*_{ij}\,\gamma^\mu W^+_\mu\,u_{Lj}.$$

Under the CP transformation, the charged gauge fields and fermion bilinears obey

$$W^-_\mu(x) \xrightarrow{CP} W^+_\mu(x'), \qquad \overline{u_{Li}}\,\gamma^\mu d_{Lj} \xrightarrow{CP} \overline{d_{Lj}}\,\gamma^\mu u_{Li}.$$

Hence the CP–transformed charged current interaction becomes[39]

$$\mathcal{L}^{CP}_{\text{kinetic,cc}}(Q_L) = \frac{g}{\sqrt{2}}\,\overline{d_{Li}}\,V_{ij}\,\gamma^\mu W^+_\mu\,u_{Lj} + \frac{g}{\sqrt{2}}\,\overline{u_{Li}}\,V^*_{ij}\,\gamma^\mu W^-_\mu\,d_{Lj}.$$

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Criterion for CP violation

In quantum field theory, a symmetry is said to be conserved if and only if the action, or equivalently the Lagrangian density, is invariant under the corresponding transformation. Therefore, under a CP transformation,

$$\mathcal{L} \xrightarrow{CP} \mathcal{L},$$

CP symmetry is conserved, whereas if

$$\mathcal{L} \xrightarrow{CP} \mathcal{L}^{CP} \neq \mathcal{L},$$

the theory violates CP symmetry. Hence, if the Lagrangian is not invariant under the CP transformation, the theory is said to violate CP symmetry. This is the precise definition of CP violation.[39]

Application to the charged current interaction

In the basis of quark mass eigenstates, the charged current part of the kinetic Lagrangian is given by

$$\mathcal{L}_{\text{kinetic,cc}} = \frac{g}{\sqrt{2}}\,\overline{u_{Li}}\,V_{ij}\,\gamma^\mu W_\mu^-\,d_{Lj} + \frac{g}{\sqrt{2}}\,\overline{d_{Li}}\,V_{ij}^*\,\gamma^\mu W_\mu^+\,u_{Lj}.$$

Under the CP transformation, the charged current becomes

$$\mathcal{L}^{CP}_{\text{kinetic,cc}} = \frac{g}{\sqrt{2}}\,\overline{d_{Li}}\,V_{ij}\,\gamma^\mu W_\mu^+\,u_{Lj} + \frac{g}{\sqrt{2}}\,\overline{u_{Li}}\,V_{ij}^*\,\gamma^\mu W_\mu^-\,d_{Lj}.$$

If the CKM matrix is real, namely if

$$V_{ij} = V_{ij}^*,$$

the CP-transformed Lagrangian coincides with the original one and CP symmetry is conserved. If, however,

$$V_{ij} \neq V_{ij}^*,$$

the two expressions differ, the Lagrangian is not invariant under CP, and CP symmetry is violated. We thus conclude that the complex nature of the CKM matrix is the origin of CP violation in the Standard Model.

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Physical meaning of complex phases

Not every complex parameter necessarily leads to CP violation. This is because one is free to perform phase redefinitions of the fermion fields,

$$\psi \longrightarrow e^{i\alpha}\,\psi,$$

which can remove many complex phases from the Lagrangian. Only those phases that cannot be eliminated by any redefinition of the fields are physically meaningful and can give rise to CP violation.

Therefore, CP is violated if and only if there exists no choice of field phases for which the Lagrangian becomes real and invariant under CP.

In the Standard Model, the Yukawa matrices are in general complex. After diagonalization of the mass matrices, one irreducible complex phase remains in the CKM matrix which cannot be removed by any rephasing of the quark fields. This single physical phase is responsible for all CP-violating effects in the quark sector.[39]

As a consequence,

$$\text{one generation} \;\Rightarrow\; \text{no CP violation,}$$

$$\text{two generations} \;\Rightarrow\; \text{no CP violation,}$$

$$\text{three generations} \;\Rightarrow\; \text{one physical CP-violating phase.}$$

In field theory language:

CP is conserved if there exists a unitary transformation of the fields such that the Lagrangian is invariant under the combined CP operation. CP is violated if no such transformation exists.

Equivalently:

CP violation occurs when the Lagrangian cannot be made real by any redefinition of the fields.

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Chapter 4

Dimensional Estimation And CP Violation

In the previous chapter we saw how the introduction of Yukawa couplings (i.e. the terms where the Higgs couples to the fermions) led to off-diagonal elements in the 3×3 matrix between the different families. By diagonalizing the 3×3 Yukawa matrix, these off-diagonal elements appear in the charged current coupling, in the Cabibbo-Kobayashi-Maskawa-matrix(CKM Matrix) V_{CKM} [34, 40]:

$$V = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}$$

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The CKM-mechanism is the origin of CP violation, and earned Kobayashi and Maskawa the Nobel prize in 2008 [8].

4.1 Properties Of CKM Matrix

1.) The CKM matrix V_{CKM} is a 3×3 **unitary** matrix [34], meaning:

$$V_{\text{CKM}}^{\dagger} V_{\text{CKM}} = V_{\text{CKM}} V_{\text{CKM}}^{\dagger} = \mathbb{I}$$

2.) The phases of the quarks can be rotated freely: $u_{Li} \to e^{i\phi_i^u} u_{Li}$ and $d_{Lj} \to e^{i\phi_j^d} d_{Lj}$ [40].
Since the overall phase is irrelevant, $2n - 1$ relative quark phases can be removed. [39]

Thus, the CKM-matrix describing the flavour couplings of n generations of up

and down type quarks has

$$2n^2 - n^2 - (2n-1) = (n-1)^2$$

free parameters. Subsequently, we can divide these free parameters into Euler angles and phases:

3.) A general $n \times n$ orthogonal matrix can be constructed from $\frac{1}{2}n(n-1)$ angles describing the rotations among the n dimensions.

4.) The remaining free parameters are the phases:

$$(n-1)^2 - \frac{1}{2}n(n-1) = \frac{1}{2}(n-1)(n-2).$$

Global Phase Redundancy and Counting of Physical CKM Parameters

In quantum field theory, a global phase transformation of a field [34],

$$\psi \to e^{i\alpha}\psi,$$

has no physical consequence. All physical observables are constructed from bilinear combinations such as

$$|\psi|^2, \quad \bar{\psi}\psi, \quad \psi^\dagger\psi,$$

which are invariant under global phase rotations. For example,

$$\bar{\psi}\psi \to (e^{-i\alpha}\bar{\psi})(e^{i\alpha}\psi) = \bar{\psi}\psi.$$

Hence, physics is completely insensitive to a common overall phase. This is called **global phase invariance**, and it implies that any such overall phase is unphysical [40].

In the Standard Model, we have n generations of up-type quarks and n generations of down-type quarks, giving a total of $2n$ quark fields. Each of these fields may be redefined by an independent phase rotation:

$$q_i \to e^{i\alpha_i} q_i.$$

This would seem to give $2n$ possible rephasings. However, if we apply the *same* phase transformation to *all* quark fields simultaneously,

$$q_i \to e^{i\theta} q_i \quad \text{for all } i,$$

then the Lagrangian remains completely unchanged. This is the trivial global phase

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rotation, which has no physical effect [8].

Therefore, among the $2n$ possible field rephasings, only $2n - 1$ are physically meaningful in removing complex phases from the CKM matrix. This global redundancy is always subtracted when counting physical parameters in the CKM matrix [40].

Step-by-Step CKM Parameter Count:

- An $n \times n$ complex matrix has $2n^2$ real parameters (since every element $z = a + bi$ has 2 real numbers).
- **The Unitary Constraint ($V^\dagger V = I$):** This forces the rows/columns to be orthonormal (length 1, orthogonal to each other) [34].
  - This imposes n^2 constraints.
  - **Remaining Free Parameters:** $2n^2 - n^2 = \mathbf{n^2}$.
- Total real parameters in an $n \times n$ unitary matrix:

$$n^2$$

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- Number of physical mixing angles (rotations between generations):

$$\frac{n(n-1)}{2}$$

- Remaining parameters are complex phases:

$$n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$$

- Removable unphysical phases from field redefinitions:

$$2n \text{ field phases} - 1 \text{ global phase} = 2n - 1$$

- **Physical phases** left after removing unphysical ones:

$$\frac{n(n+1)}{2} - (2n-1) = \frac{n^2 - 3n + 2}{2} = \frac{(n-1)(n-2)}{2}$$

Conclusion: The CKM matrix contains [8]:

- $\frac{n(n-1)}{2}$ physical mixing angles

- $\dfrac{(n-1)(n-2)}{2}$ physical CP-violating phases

Why Only 3 Generations Of quarks:

- For $n = 1$ or $n = 2$:

$$\frac{(n-1)(n-2)}{2} = 0 \quad \Rightarrow \quad \text{No CP violation}$$

- For $n = 3$:

$$\frac{(3-1)(3-2)}{2} = \frac{2 \cdot 1}{2} = 1$$

$\Rightarrow$ One irreducible physical complex phase $\Rightarrow$ CP violation is possible [23, 22].

Only when $n \geq 3$ does a physical CP-violating phase survive field rephasings, making CP violation possible in the Standard Model [23].

Let us now look at the consequences of the unitarity condition for the CKM-matrix [40]:

$$V^{\dagger}V = VV^{\dagger} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} V_{ud}^* & V_{cd}^* & V_{td}^* \\ V_{us}^* & V_{cs}^* & V_{ts}^* \\ V_{ub}^* & V_{cb}^* & V_{tb}^* \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(4.1)

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By doing matrix multiplication, first with row by column with same indices, then differently, we get:

$$\begin{aligned} V_{ud}V_{ud}^* + V_{us}V_{us}^* + V_{ub}V_{ub}^* &= 1 \\ V_{cd}V_{cd}^* + V_{cs}V_{cs}^* + V_{cb}V_{cb}^* &= 1 \\ V_{td}V_{td}^* + V_{ts}V_{ts}^* + V_{tb}V_{tb}^* &= 1 \end{aligned}$$

(4.2)

The remaining relations are known as the orthogonality conditions [8, 40]:

$$
\begin{aligned}
V_{ud}V_{cd}^* + V_{us}V_{cs}^* + V_{ub}V_{cb}^* &= 0 \\
V_{ud}V_{td}^* + V_{us}V_{ts}^* + V_{ub}V_{tb}^* &= 0 \\
V_{cd}V_{ud}^* + V_{cs}V_{us}^* + V_{cb}V_{ub}^* &= 0 \\
V_{cd}V_{td}^* + V_{cs}V_{ts}^* + V_{cb}V_{tb}^* &= 0 \\
V_{td}V_{ud}^* + V_{ts}V_{us}^* + V_{tb}V_{ub}^* &= 0 \\
V_{td}V_{cd}^* + V_{ts}V_{cs}^* + V_{tb}V_{cb}^* &= 0
\end{aligned}
\tag{4.3}
$$

Three of the six equations are simply the complex conjugate version. An additional three interesting equations arise from the unitarity relation $V^\dagger V = 1$:

$$
\begin{aligned}
V_{ud}^*V_{us} + V_{cd}^*V_{cs} + V_{td}^*V_{ts} &= 0 \\
V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* &= 0 \\
V_{us}^*V_{ud} + V_{cs}^*V_{cd} + V_{ts}^*V_{td} &= 0 \\
V_{us}V_{ub}^* + V_{cs}V_{cb}^* + V_{ts}V_{tb}^* &= 0 \\
V_{ub}^*V_{ud} + V_{cb}^*V_{cd} + V_{tb}^*V_{td} &= 0 \\
V_{ub}^*V_{us} + V_{cb}^*V_{cs} + V_{tb}^*V_{ts} &= 0
\end{aligned}
\tag{4.4}
$$

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Geometrical Interpretation of CKM Unitarity Relations

From the unitarity of the CKM matrix V, one of the relations we obtain is [40]:

$$V_{ud}^*V_{ub} + V_{cd}^*V_{cb} + V_{td}^*V_{tb} = 0$$

This is a sum of three complex numbers. Since their total is zero, we can interpret them geometrically.

Triangle in the Complex Plane

In complex analysis:

- If $z_1 + z_2 + z_3 = 0$, where each $z_i \in \mathbb{C}$, then the three complex numbers can be treated as vectors in the complex plane.
- These vectors can be arranged tip-to-tail to form a closed triangle.

This is because, in two-dimensional geometry, a set of vectors that sum to zero forms a closed loop — in this case, a triangle [8].

Physical Meaning

- The unitarity triangle is a geometric representation of CKM unitarity conditions where complex phases are involved [40].
- The area of this triangle is proportional to the amount of **CP violation** in the Standard Model [23, 22].
- If all CKM elements were real, the triangle would collapse to a line (area zero), and there would be no CP violation [5].

Thus, these triangle relations provide a powerful visual tool to study CP-violating effects in the quark sector [8].

4.2 CKM Matrix Parameterization

The **parameterization of a matrix** refers to expressing the elements of that matrix in terms of a set of physical parameters—in this case, three mixing angles and one complex phase that governs CP violation [34, 40]. This ensures all physically meaningful content is preserved and represented.

We want to describe the 3×3 CKM matrix using a minimal set of physical parameters:

- Three mixing angles: $\theta_{12}, \theta_{13}, \theta_{23}$
- One complex phase: δ_{13} (responsible for CP violation)

We define:

$$c_{ij} = \cos\theta_{ij},$$
$$s_{ij} = \sin\theta_{ij}.$$

Construction of the CKM Matrix

We construct the full CKM matrix as the product of three basic rotations in generation space [8]:

$$V_{\text{CKM}} = R_{23} \cdot R_{13} \cdot R_{12}$$

- R_{12} rotates the 1st and 2nd generations:

$$R_{12} = \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- R_{13} rotates the 1st and 3rd generations and includes the complex phase δ_{13}:

$$R_{13} = \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta_{13}} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta_{13}} & 0 & c_{13} \end{pmatrix}$$

- R_{23} rotates the 2nd and 3rd generations:

$$R_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}$$

Final CKM Matrix

Multiplying these three matrices, we get the standard CKM matrix [8]:

$$V_{\text{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{13}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{13}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{13}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{13}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{13}} & c_{23}c_{13} \end{pmatrix}$$

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Importance:

- The CKM matrix encodes how quark flavor states mix under weak interactions [34].
- The complex phase δ_{13} introduces CP violation in the Standard Model [23].
- The matrix elements are constrained by experimental data and help describe phenomena such as decay rates and CP-violating processes [38, 10].

The phase can be made to appear in many elements, and is chosen here to appear in the matrix describing the relation between the 1st and 3rd family [40].

4.3 The Jarlskog Invariant and CP Violation

CP violation in the Standard Model is encoded in the complex phase of the CKM matrix [23]. A necessary condition for CP violation is that at least one of the combinations $V_{ij}V_{kl} \neq V_{ij}^*V_{kl}^*$, which translates to the requirement that the unitarity triangle formed in the complex plane has a non-zero area [22].

This area is directly related to a quantity known as the **Jarlskog invariant**, denoted by J [23]. All six unitarity triangles (from different orthogonality conditions of the CKM

matrix rows and columns) are found to have the same area, and hence the same value of J [22, 28]. It is defined as the imaginary part of a specific product of CKM matrix elements:

$$J = \mathrm{Im}(V_{11}V_{22}V_{12}^*V_{21}^*) = \mathrm{Im}(V_{22}V_{33}V_{23}^*V_{32}^*) = \cdots \tag{4.5}$$

There are in total nine such combinations, and all give the same value of J [23, 5].

Derivation of the Jarlskog Invariant J

The Jarlskog invariant J quantifies CP violation in the Standard Model [23, 22] and is defined as:

$$J = \mathrm{Im}(V_{ud}V_{cs}V_{us}^*V_{cd}^*)$$

We will compute this using the standard CKM parameterization [8]:

$$V_{\mathrm{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{13}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{13}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{13}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{13}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{13}} & c_{23}c_{13} \end{pmatrix}$$

where the relevant matrix elements are:

$$\begin{aligned} V_{ud} &= c_{12}c_{13}, \\ V_{us} &= s_{12}c_{13}, \quad \text{so } V_{us}^* = s_{12}c_{13}, \\ V_{cs} &= c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta}, \\ V_{cd} &= -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta}, \quad \text{so } V_{cd}^* = -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{-i\delta}. \end{aligned}$$

Substitute these expressions into the definition of J [23]:

$$J = \mathrm{Im}\left[V_{ud} \cdot V_{cs} \cdot V_{us}^* \cdot V_{cd}^*\right]$$

$$= \mathrm{Im}\left[(c_{12}c_{13})(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta})(s_{12}c_{13})(-s_{12}c_{23} - c_{12}s_{23}s_{13}e^{-i\delta})\right]$$

Group constants:

$$= c_{12}s_{12}c_{13}^2 \cdot \mathrm{Im}\left[(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta})(-s_{12}c_{23} - c_{12}s_{23}s_{13}e^{-i\delta})\right]$$

Now expand the product inside the imaginary part:

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$$
\begin{aligned}
&(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta}) \cdot (-s_{12}c_{23} - c_{12}s_{23}s_{13}e^{-i\delta}) \\
&= c_{12}c_{23} \cdot (-s_{12}c_{23}) \\
&\quad + c_{12}c_{23} \cdot (-c_{12}s_{23}s_{13}e^{-i\delta}) \\
&\quad - s_{12}s_{23}s_{13}e^{i\delta} \cdot (-s_{12}c_{23}) \\
&\quad - s_{12}s_{23}s_{13}e^{i\delta} \cdot (-c_{12}s_{23}s_{13}e^{-i\delta}) \\
&= -c_{12}s_{12}c_{23}^2 \\
&\quad - c_{12}^2 c_{23}s_{23}s_{13}e^{-i\delta} \\
&\quad + s_{12}^2 c_{23}s_{23}s_{13}e^{i\delta} \\
&\quad + c_{12}s_{12}s_{23}^2 s_{13}^2
\end{aligned}
$$

Take the imaginary part:

$$\mathrm{Im} = \mathrm{Im}\left[-c_{12}^2 c_{23}s_{23}s_{13}e^{-i\delta} + s_{12}^2 c_{23}s_{23}s_{13}e^{i\delta}\right]$$

Using $\mathrm{Im}(ae^{i\delta} + be^{-i\delta}) = (a - b)\sin\delta$, we get:

$$\mathrm{Im} = (c_{12}^2 + s_{12}^2)c_{23}s_{23}s_{13}\sin\delta$$

Since $c_{12}^2 + s_{12}^2 = 1$, we obtain:

$$\mathrm{Im} = c_{23}s_{23}s_{13}\sin\delta$$

Thus,

$$J = c_{12}s_{12}c_{13}^2 \cdot c_{23}s_{23}s_{13}\sin\delta$$

$$\boxed{J = c_{12}c_{13}^2 c_{23}s_{12}s_{13}s_{23}\sin\delta}$$

This is the full expression for the Jarlskog invariant derived explicitly from the standard CKM parameterization [23, 40].

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Derivation of the Jarlskog Determinant

Notation and Setup

We work in the weak basis where the up-type quark mass-squared matrix is diagonal [28]:

$$H_u \equiv M_u M_u^\dagger = \mathrm{diag}(m_u^2, m_c^2, m_t^2). \tag{4.6}$$

In this basis, the down-type mass-squared matrix is [22]

$$H_d \equiv M_d M_d^\dagger = V\,\mathrm{diag}(m_d^2, m_s^2, m_b^2)\,V^\dagger, \tag{4.7}$$

where V is the CKM matrix.

Define the commutator

$$C \equiv [H_u, H_d] = H_u H_d - H_d H_u. \tag{4.8}$$

This commutator C measures the non-commutativity between the up-type and down-type mass matrices. Physically, it encodes the incompatibility of their diagonalization, and is central to CP violation in the Standard Model [23, 22].

Since H_u is diagonal, its matrix elements satisfy

$$(H_u)_{ij} = m_{u_i}^2 \delta_{ij}.$$

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Therefore, for $i \neq j$,

$$\begin{aligned} C_{ij} &= (H_u)_{ii}(H_d)_{ij} - (H_d)_{ij}(H_u)_{jj} \\ &= (m_{u_i}^2 - m_{u_j}^2)(H_d)_{ij} \equiv \Delta_{ij}^{(u)}(H_d)_{ij}, \end{aligned} \tag{4.9}$$

while $C_{ii} = 0$. Thus C is purely off-diagonal [28].

Explicitly, the commutator matrix C takes the form:

$$C = \begin{pmatrix} 0 & \Delta_{12}^{(u)}(H_d)_{12} & \Delta_{13}^{(u)}(H_d)_{13} \\ \Delta_{21}^{(u)}(H_d)_{21} & 0 & \Delta_{23}^{(u)}(H_d)_{23} \\ \Delta_{31}^{(u)}(H_d)_{31} & \Delta_{32}^{(u)}(H_d)_{32} & 0 \end{pmatrix}$$

This matrix is anti-Hermitian ($C^\dagger = -C$), and its structure reflects the hierarchical nature of the quark masses and the flavor-changing structure of the CKM matrix [3]. The off-diagonal elements depend on both mass differences of up-type quarks and the non-diagonal structure of H_d.

Consider the 3×3 commutator matrix C with vanishing diagonal elements ($C_{ii} = 0$):

$$C = \begin{pmatrix} 0 & C_{12} & C_{13} \\ C_{21} & 0 & C_{23} \\ C_{31} & C_{32} & 0 \end{pmatrix}. \tag{4.10}$$

Using the standard Leibniz formula or cofactor expansion along the first row [28]:

$$\begin{aligned} \det C &= 0 \cdot (0 - C_{23}C_{32}) - C_{12} \cdot (C_{21} \cdot 0 - C_{23}C_{31}) + C_{13} \cdot (C_{21}C_{32} - 0 \cdot C_{31}) \\ &= 0 + C_{12}C_{23}C_{31} + C_{13}C_{21}C_{32} \\ &= C_{12}C_{23}C_{31} + C_{32}C_{21}C_{13}. \end{aligned}$$

The determinant is the sum of two cyclic products:

$$\det C = \underbrace{C_{12}C_{23}C_{31}}_{\text{Term 1}} + \underbrace{C_{13}C_{21}C_{32}}_{\text{Term 2}}. \tag{4.11}$$

Recall the definition of the commutator elements in the basis where H_u is diagonal [22]:

$$C_{ij} = (m^2_{u_i} - m^2_{u_j})(H_d)_{ij} \equiv \Delta^{(u)}_{ij}(H_d)_{ij}. \tag{4.12}$$

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Substitute this into **Term 1**:

$$\begin{aligned} \text{Term 1} &= \left[\Delta^{(u)}_{12}(H_d)_{12}\right] \left[\Delta^{(u)}_{23}(H_d)_{23}\right] \left[\Delta^{(u)}_{31}(H_d)_{31}\right] \\ &= \Delta^{(u)}_{12}\Delta^{(u)}_{23}\Delta^{(u)}_{31} \left[(H_d)_{12}(H_d)_{23}(H_d)_{31}\right]. \end{aligned}$$

Substitute into **Term 2**:

$$\begin{aligned} \text{Term 2} &= \left[\Delta^{(u)}_{13}(H_d)_{13}\right] \left[\Delta^{(u)}_{21}(H_d)_{21}\right] \left[\Delta^{(u)}_{32}(H_d)_{32}\right] \\ &= \Delta^{(u)}_{13}\Delta^{(u)}_{21}\Delta^{(u)}_{32} \left[(H_d)_{13}(H_d)_{21}(H_d)_{32}\right]. \end{aligned}$$

The mass-squared difference is antisymmetric: $\Delta^{(u)}_{ij} = -\Delta^{(u)}_{ji}$. We rewrite the scalar prefactor of **Term 2** to match **Term 1**:

$$\begin{aligned} \Delta^{(u)}_{13} &= -\Delta^{(u)}_{31} \\ \Delta^{(u)}_{21} &= -\Delta^{(u)}_{12} \\ \Delta^{(u)}_{32} &= -\Delta^{(u)}_{23}. \end{aligned}$$

Multiplying these gives:

$$\Delta_{13}^{(u)}\Delta_{21}^{(u)}\Delta_{32}^{(u)} = (-1)(-1)(-1)\Delta_{31}^{(u)}\Delta_{12}^{(u)}\Delta_{23}^{(u)} = -\Delta_{12}^{(u)}\Delta_{23}^{(u)}\Delta_{31}^{(u)}. \tag{4.13}$$

Since H_d is Hermitian, $(H_d)_{ji} = (H_d)_{ij}^*$. We rewrite the matrix product of **Term 2**:

$$(H_d)_{13}(H_d)_{21}(H_d)_{32} = (H_d)_{31}^*(H_d)_{12}^*(H_d)_{23}^* = \left[(H_d)_{12}(H_d)_{23}(H_d)_{31}\right]^*. \tag{4.14}$$

Let $P = \Delta_{12}^{(u)}\Delta_{23}^{(u)}\Delta_{31}^{(u)}$ and $Z = (H_d)_{12}(H_d)_{23}(H_d)_{31}$.
From equation 2.7, the determinant is the sum of Term 1 and Term 2:

$$\begin{aligned}\det C &= (\text{Term 1}) + (\text{Term 2})\\ &= P \cdot Z + (-P) \cdot Z^*\\ &= P(Z - Z^*).\end{aligned}$$

Using the complex identity $Z - Z^* = 2i\,\mathrm{Im}(Z)$:

$$\det C = P \cdot [2i\,\mathrm{Im}(Z)]. \tag{4.15}$$

Substituting P and Z back gives the final result (Eq 2.9):

$$\boxed{\det C = 2i\,\Delta_{12}^{(u)}\Delta_{23}^{(u)}\Delta_{31}^{(u)}\ \mathrm{Im}\left[(H_d)_{12}(H_d)_{23}(H_d)_{31}\right]} \tag{4.16}$$

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Triple Product in Terms of CKM Matrix

Using [23]

$$(H_d)_{ij} = \sum_{k=d,s,b} V_{ik}\, m_k^2\, V_{jk}^*, \tag{4.17}$$

the triple product becomes

$$(H_d)_{12}(H_d)_{23}(H_d)_{31} = \sum_{k,\ell,m} V_{1k}V_{2k}^*\, V_{2\ell}V_{3\ell}^*\, V_{3m}V_{1m}^*\, m_k^2 m_\ell^2 m_m^2. \tag{4.18}$$

The formula for $(H_d)_{ij}$ arises from the diagonalization of the Hermitian mass matrix [34].

Let D^2 be the diagonal matrix of the physical squared masses for the down-type quarks:

$$D^2 = \mathrm{diag}(m_d^2, m_s^2, m_b^2). \tag{4.19}$$

In the Standard Model, the mass matrix in the flavor (weak) basis, H_d, is related to the physical mass basis D^2 by a unitary rotation. In the basis where up-type quarks are

already diagonal, this rotation is given exactly by the CKM matrix V [34, 22]:

$$H_d = VD^2V^\dagger. \tag{4.20}$$

We write this matrix product explicitly using index notation. The element $(H_d)_{ij}$ is given by

$$(H_d)_{ij} = \sum_k V_{ik}(D^2V^\dagger)_{kj}. \tag{4.21}$$

First, evaluate the term $(D^2V^\dagger)_{kj}$. Since D^2 is diagonal, its elements are $(D^2)_{kl} = m_k^2\delta_{kl}$. Thus,

$$(D^2V^\dagger)_{kj} = \sum_l (D^2)_{kl}(V^\dagger)_{lj} = \sum_l (m_k^2\delta_{kl})(V_{jl}^*) = m_k^2 V_{jk}^*. \tag{4.22}$$

Substituting this back into the expression for $(H_d)_{ij}$, we obtain the spectral decomposition:

$$\boxed{(H_d)_{ij} = \sum_{k=d,s,b} V_{ik}\, m_k^2\, V_{jk}^*.} \tag{4.23}$$

Physically, this represents:

- V_{ik}: Rotation from the weak basis i to the mass eigenstate k.
- m_k^2: The eigenvalue (mass squared) of state k.
- V_{jk}^*: Rotation back from mass eigenstate k to weak basis j.

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Terms containing repeated indices ($k = \ell$, etc.) are manifestly real and do not contribute to the imaginary part. Hence only terms with k, ℓ, m all distinct contribute [28].

Using CKM unitarity, the imaginary part of the CKM product factorizes universally [23, 22]:

$$\mathrm{Im}(V_{1k}V_{2k}^*V_{2\ell}V_{3\ell}^*V_{3m}V_{1m}^*) = J\,\varepsilon_{k\ell m}, \tag{4.24}$$

where J is the Jarlskog invariant and $\varepsilon_{k\ell m}$ is the Levi–Civita symbol over $\{d, s, b\}$.

Therefore,

$$\mathrm{Im}[(H_d)_{12}(H_d)_{23}(H_d)_{31}] = J \sum_{k,\ell,m} \varepsilon_{k\ell m} m_k^2 m_\ell^2 m_m^2. \tag{4.25}$$

The sum evaluates to the Vandermonde determinant [28]:

$$\sum_{k,\ell,m} \varepsilon_{k\ell m} m_k^2 m_\ell^2 m_m^2 = (m_b^2 - m_s^2)(m_b^2 - m_d^2)(m_s^2 - m_d^2), \tag{4.26}$$

where k, ℓ, m are all distinct.

Hence,

$$\boxed{\mathrm{Im}[(H_d)_{12}(H_d)_{23}(H_d)_{31}] = J\,(m_b^2 - m_s^2)(m_b^2 - m_d^2)(m_s^2 - m_d^2).} \tag{4.27}$$

Jarlskog Determinant Formula

Substituting into the determinant expression yields [28]

$$\boxed{\det C = 2i\,J \prod_{i<j}(m_{u_i}^2 - m_{u_j}^2) \prod_{a<b}(m_{d_a}^2 - m_{d_b}^2).} \tag{4.28}$$

This expression vanishes if any two quarks of the same charge are degenerate, demonstrating that CP violation requires both a physical CKM phase and non-degenerate quark masses [23, 3].

4.3.1 Relation to CKM Matrix Elements

To evaluate the Jarlskog invariant J, we relate the mixing parameters s_{ij} and c_{ij} to the magnitudes of the CKM matrix elements $|V_{ij}|$. In the standard parameterization, the CKM matrix is given by [8]:

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$$V_{\mathrm{CKM}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix},$$

where $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$, and δ is the phase responsible for all CP-violating phenomena in flavor-changing processes in the SM [38]. The angles θ_{ij} can be chosen to lie in the first quadrant, so $s_{ij}, c_{ij} \geq 0$.

It is known experimentally that $s_{13} \ll s_{23} \ll s_{12} \ll 1$, and it is convenient to exhibit this hierarchy using the Wolfenstein parameterization [8, 40]. We define:

$$s_{12} = \lambda = \frac{|V_{us}|}{\sqrt{|V_{ud}|^2 + |V_{us}|^2}}, \quad s_{23} = A\lambda^2 = \lambda \left|\frac{V_{cb}}{V_{us}}\right|,$$

$$s_{13}e^{i\delta} = V_{ub}^* = A\lambda^3(\rho + i\eta)$$

These relations ensure that $\bar{\rho} + i\bar{\eta} = -(V_{ud}V_{ub}^*)/(V_{cd}V_{cb}^*)$ is phase convention independent, and the CKM matrix written in terms of λ, A, $\bar{\rho}$, and $\bar{\eta}$ is unitary to all orders in λ [8]. The definitions of $\bar{\rho}$, $\bar{\eta}$ reproduce all approximate results in the literature; i.e.,

$\bar{\rho} = \rho(1 - \lambda^2/2 + \ldots)$ and $\bar{\eta} = \eta(1 - \lambda^2/2 + \ldots)$, and one can write V_{CKM} to $\mathcal{O}(\lambda^4)$ either in terms of $\bar{\rho}$, $\bar{\eta}$ or, traditionally [23]

$$V_{\text{CKM}} = \begin{pmatrix} 1 - \lambda^2/2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \lambda^2/2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4).$$

From this structure, we can directly identify the mixing angles with specific matrix elements. Since experimentally $|V_{ub}| \ll |V_{cb}| \ll |V_{us}|$, we know the angles are small. Specifically, because $s_{13} \approx 0$, we can approximate $c_{13} \approx 1$. This yields the following direct relations [8]:

$$s_{13} = |V_{ub}|, \tag{4.29}$$

$$s_{12} = \frac{|V_{us}|}{c_{13}} \approx |V_{us}|, \tag{4.30}$$

$$s_{23} = \frac{|V_{cb}|}{c_{13}} \approx |V_{cb}|. \tag{4.31}$$

Using the standard values for the CKM elements [8]:

- $|V_{us}| \approx 0.2245 \implies s_{12} \approx 0.2245$ (and $c_{12} = \sqrt{1 - s_{12}^2} \approx 0.9745$),
- $|V_{cb}| \approx 0.0410 \implies s_{23} \approx 0.0410$,
- $|V_{ub}| \approx 0.0037 \implies s_{13} \approx 0.0037$.

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From above, we have:

$$V_{ub} = s_{13} e^{-i\delta}.$$

In the Wolfenstein parameterization [8, 40],

$$V_{ub} = A\lambda^3(\rho - i\eta).$$

Equating the two expressions gives

$$s_{13} e^{-i\delta} = A\lambda^3(\rho - i\eta).$$

Taking the modulus of both sides,

$$s_{13} = |V_{ub}| = A\lambda^3\sqrt{\rho^2 + \eta^2}.$$

Dividing both sides by the magnitude,

$$e^{-i\delta} = \frac{\rho - i\eta}{\sqrt{\rho^2 + \eta^2}}.$$

Comparing with the general complex representation

$$e^{-i\delta} = \cos\delta - i\sin\delta,$$

we identify

$$\cos\delta = \frac{\rho}{\sqrt{\rho^2+\eta^2}}, \qquad \sin\delta = \frac{\eta}{\sqrt{\rho^2+\eta^2}}.$$

Using the experimentally determined Wolfenstein parameters [8]

$$\bar{\rho} \approx 0.145, \qquad \bar{\eta} \approx 0.355,$$

and the relation [23]

$$\rho = \frac{\bar{\rho}}{1-\lambda^2/2}, \qquad \eta = \frac{\bar{\eta}}{1-\lambda^2/2},$$

with $\lambda = 0.2245$, we first compute

$$1 - \frac{\lambda^2}{2} \approx 0.9748.$$

Thus,

$$\rho \approx \frac{0.145}{0.9748} \approx 0.149,$$
$$\eta \approx \frac{0.355}{0.9748} \approx 0.364.$$

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Now,

$$\sqrt{\rho^2+\eta^2} = \sqrt{(0.149)^2+(0.364)^2} = \sqrt{0.0222+0.1325} = \sqrt{0.1547} \approx 0.393.$$

Therefore,

$$\sin\delta = \frac{0.364}{0.393} \approx 0.93.$$

Hence,

$$\boxed{\sin\delta \approx 0.93}$$

4.3.2 Numerical Evaluation of J

The Jarlskog invariant is defined in the standard parameterization as [23]:

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13}\sin\delta. \tag{4.32}$$

Applying the small angle approximations $c_{13} \approx 1$ and $c_{23} \approx 1$, this simplifies to [40]:

$$\begin{aligned} J &\approx c_{12}s_{12}s_{23}s_{13}\sin\delta \\ &\approx (0.9745)(0.2245)(0.0410)(0.0037)(0.93) \\ &\approx 3.0 \times 10^{-5}. \end{aligned} \tag{4.33}$$

This value represents the maximum strength of CP violation provided by the Standard Model quark sector [23, 10].

Dimensional Analysis

We need a dimensionless parameter δ_{CP} to compare with the baryon asymmetry ($\eta \sim 10^{-10}$) [39].

- Our determinant has units of $[\text{Mass}]^{12}$. (Count them: 6 mass-squared terms = 12 powers of mass).
- To make it dimensionless, we must divide by the temperature scale to the power of 12.

The CP Violating calculation Formula [18, 19, 38]:

$$\delta_{CP} \approx J\frac{(m_t^2 - m_c^2)(m_t^2 - m_u^2)(m_c^2 - m_u^2)(m_b^2 - m_s^2)(m_b^2 - m_d^2)(m_s^2 - m_d^2)}{T_{EW}^{12}} \tag{4.34}$$

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We plug in the numbers to see why the Standard Model fails [16, 18]:

- $T \approx 100$ GeV
- $J \approx 3 \times 10^{-5}$

Now using the quarks' masses and electroweak scale temperature [8],

$$m_u = 0.0022\text{ GeV}, \quad m_d = 0.0047\text{ GeV}, \quad m_s = 0.096\text{ GeV}$$

$$m_c = 1.27\text{ GeV}, \quad m_b = 4.18\text{ GeV}, \quad m_t = 172.9\text{ GeV}$$

$$T_{EW} = 100\text{ GeV}, \quad J = 3.0 \times 10^{-5}$$

For Up-type product:

$$\Pi_{\text{up}} = (m_t^2 - m_c^2)(m_t^2 - m_u^2)(m_c^2 - m_u^2)$$

$$\Pi_{\text{up}} \approx 2.98927971 \times 10^4 \times 2.989441 \times 10^4 \times 1.61289516$$

$$\Pi_{\text{up}} \approx 1.4413 \times 10^9$$

For Down-type product:

$$\Pi_{\text{down}} = (m_b^2 - m_s^2)(m_b^2 - m_d^2)(m_s^2 - m_d^2)$$

$$\Pi_{\text{down}} \approx 17.463184 \times 17.47237791 \times 9.19391 \times 10^{-3}$$

$$\Pi_{\text{down}} \approx 2.8053$$

so, now we have:

$$\Pi_{\text{total}} = \Pi_{\text{up}}\Pi_{\text{down}} = (1.4413 \times 10^9) \times (2.8053)$$

$$\Pi_{\text{total}} \approx 4.0433 \times 10^9$$

Using these values, we get [19]:

$$\delta_{CP} = J\frac{\Pi_{\text{total}}}{T_{EW}^{12}} = 3.0 \times 10^{-5} \times \frac{4.0433 \times 10^9}{10^{24}}$$

$$\delta_{CP} = 3.0 \times 10^{-5} \times 4.0433 \times 10^{-15}$$

$$\delta_{CP} \approx 1.21 \times 10^{-19}$$

$$\boxed{\delta_{CP} \approx 1.2 \times 10^{-19}}$$

This is why the formula exists: It proves mathematically that because the **Up, Down, and Strange quarks are so light** compared to the 100 GeV temperature of the early universe, the Standard Model's CP violation is washed out and becomes irrelevant [21, 19].

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4.4 Important Note

It is crucial to distinguish this quantity from the actual baryon-to-entropy ratio. The value $D_{CP} \sim 10^{-19}$ is a **dimensional estimate** of the maximal CP-violating strength using Jarlskog invariant [23], not a rigorous calculation of n_B/s.

The actual baryon-to-entropy ratio Y_B is defined as [39]:

$$Y_B = \frac{n_B - n_{\bar{B}}}{s} \tag{4.35}$$

where $s \sim g_* T^3$ is the entropy density. A full calculation of Y_B requires solving Boltzmann equations to account for the sphaleron rate, expansion of the universe, and washout effects. Such a calculation would likely yield a value even smaller than the dimensional estimate.

However, the conclusion remains robust: since the dimensional upper bound (10^{-19}) is already ten orders of magnitude smaller than the observed asymmetry ($\eta \sim 10^{-10}$), we can definitively state that **Standard Model CP violation is insufficient** [16, 18, 21] to explain the matter content of the universe. New sources of CP violation (Physics Beyond the Standard Model) are required .

Strictly speaking, the number $\delta_{CP} \sim 10^{-19}$ is not the baryon-to-entropy ratio (n_B/s). It is a dimensionless measure of the strength of CP violation at the electroweak scale [3].

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Conclusions

This thesis report examined the baryon asymmetry of the universe using the Standard Model through two complementary quantitative methods. The first method was a direct computation of the baryon-to-entropy ratio using the electroweak baryogenesis formalism [10, 39]. The second method involved an independent dimensional estimate of the CP-violating strength through the Jarlskog determinant [28, 24].

The electroweak baryogenesis calculation was conducted at $T = 100$ GeV using the full Standard Model particle content. This included $N_b = 28$, $N_f = 90$, and $g_* = 106.75$. The analytical result produced $\eta_1 \approx -1.89 \times 10^{-15}$. In contrast, the Python implementation, which performed the full complex CKM contraction $\beta_k = (UBU^\dagger)_{kk}$ without approximation [8], yielded:

$$\eta_1 = \frac{n_B}{s} \approx -1.253 \times 10^{-33}, \tag{4.36}$$

This value is twenty-three orders of magnitude lower than the observed value $\eta_{\text{obs}} \approx 8.7 \times 10^{-11}$. The computed-to-observed ratio is 1.440×10^{-23}.

Separately, the Jarlskog invariant $J \approx 3.0 \times 10^{-5}$ [40], combined with the full quark mass-squared difference product, provides:

$$\delta_{CP} \approx J \cdot \frac{\prod_{i<j}(m_{u_i}^2 - m_{u_j}^2) \prod_{a<b}(m_{d_a}^2 - m_{d_b}^2)}{T_{\text{EW}}^{12}} \approx 1.2 \times 10^{-19}, \tag{4.37}$$

This result is ten orders of magnitude below the observed asymmetry. Both findings are fully consistent. The CKM hierarchy, as well as the lightness of the u, d, and s quarks compared to T_{EW}, along with Boltzmann suppression, all contribute to making Standard Model CP violation inadequate by many orders of magnitude.

Moreover, for $m_h \approx 125$ GeV, the electroweak phase transition is a smooth crossover rather than a strong first-order transition [26, 11, 32]. This indicates that the third Sakharov condition [37] is also not met within the Standard Model. Therefore, the conclusion is clear: new physics beyond the Standard Model is not just desirable but necessary. Additional CP-violating phases [**grojean2005**], strongly first-order transitions, or leptogenesis through heavy Majorana neutrinos [17, 13] must exist to explain the observed matter-antimatter asymmetry of the universe.

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